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Highlights

- Ternary electrolyte mixtures consisting of lithium salt and non-electrolyte solvents
- Eigenvalues of second-order diffusivity matrix by experiment and simulation
- Molecular simulations give access to the influence of intermolecular interactions
- Ion dissociation by the solvent can lead to negative diffusion coefficient
- Slowing down of diffusive mass transport is related to the formation of ion pairs

Diffusion in Ternary Electrolyte Mixtures by Dynamic Light Scattering and Molecular Dynamics Simulations

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ABSTRACT

This work contributes to the development of an understanding of diffusive mass transport in ternary electrolyte mixtures of technical relevance. For this, ternary electrolyte mixtures containing lithium bis(trifluoromethylsulfonyl)imide ([Li][NTf_2]) dissolved in binary mixtures of acetonitrile (ACN) with either dimethyl carbonate (DMC) or dimethyl sulfoxide (DMSO) are characterized by determining mass diffusivities using dynamic light scattering (DLS) and equilibrium molecular dynamics (MD) simulations over a broad range of temperatures, as well as solute and solvent compositions. By DLS, the eigenvalues of the second-order Fick diffusion coefficient matrix $\boldsymbol{D}$ could be accessed with an average expanded experimental uncertainty (coverage factor $k = 2$) of 11%. For the first time, the methodology for accessing Fick diffusion coefficients D_{ij} of ternary electrolyte mixtures in the amount-averaged reference frame by MD simulations is presented. For this, Maxwell-Stefan (MS) diffusion coefficients $Đ_{ij}$ and thermodynamic factors Γ_{ij} are obtained separately and combined to obtain D_{ij}. With the help of the eigenvalues of $\boldsymbol{D}$ from DLS, the MD simulations and the employed force fields could be validated. By combining the results from DLS and MD simulations, the influence of the intermolecular interactions on the diffusive mass transport could be studied. For mixtures based on ACN and DMC, for example, negative off-diagonal elements of $\boldsymbol{D}$ could be linked to a diffusion of ACN molecules towards lithium cations due to the involvement in the dissociation of the ions. These observations were further corroborated by MD simulations by studying the composition of the solvation shell surrounding the cations.

KEYWORDS

Ternary Electrolyte Mixtures, Fick Diffusion Coefficient, Maxwell-Stefan Diffusion Coefficient, Dynamic Light Scattering, Molecular Dynamics Simulations

1 Introduction

Ternary electrolyte mixtures are the working fluids in many areas of chemical and energy engineering [1–5]. A ternary electrolyte mixture, such as used in lithium-ion batteries, typically consists of a salt dissolved in a binary mixture of molecular solvents [6]. Here, one solvent with a large dielectric constant ε_r is usually used to ensure good ion dissociation, while a co-solvent with a relatively low viscosity η is used to enhance the fluidity of the mixture with the intention of improving ion transport [7]. Typical solvents with a larger ε_r include dimethyl sulfoxide (DMSO), acetonitrile (ACN) or ethylene carbonate (EC) [5,8,9], while linear carbonates such as dimethyl carbonate (DMC) or diethyl carbonate (DEC) [5,9] are commonly used as low-η solvents. The transport properties of the electrolyte mixtures are indispensable for the design and optimization of related applications [6,10,11] and the development of next-generation technologies such as extreme-fast-charging batteries [12]. In this context, Fick diffusion coefficients D_{ij}, describing the diffusive mass transport due to applied concentration gradient in an isothermal isobaric system, are of particular interest as they limit the performance of electrolytic applications [6,11–13]. Despite their importance, available data for D_{ij} in ternary electrolyte mixtures are scarce in the literature [12]. In this connection, predictive models may be considered as an alternative approach. Although several predictive models have been proposed in the literature for binary electrolyte mixtures [14,15], such as the Nernst–Haskell equation [15], which relies on the conductance of the constituent ions, their applicability for estimating diffusion coefficients in ternary electrolyte mixtures has not been documented. The testing, validation, and further development of such models still rely on accurate experimental diffusivity measurements.

For ternary electrolyte mixtures consisting of a salt dissolved in a binary mixture of molecular solvents, which are of interest for the present work, three different components can be identified. Due to the dissociation of the electrolyte into anions and cations, four different species are present. This means that diffusive fluxes for four species can be defined. Due to the electroneutrality condition [16–18], the movements of the ionic species are coupled, which reduces the number of independent fluxes by one [19]. The closing condition, which couples the fluxes of all components on the basis of the mass conservation, further reduces the number of independent fluxes by one. This means that for the description of the diffusive mass transport for the ternary electrolyte mixtures at hand, a second-order Fick diffusion coefficient matrix $\boldsymbol{D}$ is required [20]. Unlike binary mixtures, the diffusivities of ternary mixtures depend on the reference frame used to determine them.

Various experimental techniques, such as Taylor dispersion [21], Gouy interferometry [22–24], and the Stocks diaphragm cell [22], have been used in the literature to determine the diffusion coefficient in electrolyte mixtures. An overview of these techniques can be found in the textbook of Cussler [25]. These techniques rely on the application of macroscopic concentration gradients. In contrast, dynamic light scattering (DLS), applied in the present work, determines mass diffusion coefficients by analyzing microscopic concentration fluctuations at macroscopic thermodynamic equilibrium in an absolute manner, eliminating the need for calibration [26]. The fluctuations investigated by DLS are caused by the thermal motion of molecules. This means that no macroscopic gradients need to be applied. This reduces both experimental time and errors caused by disruptive effects, such as advection [27]. For an n-component mixture, DLS can give access to n–1 individual mass modes [26,27], according to Onsager's regression hypothesis [28,29]. These mass modes reflect the eigenvalues of the $(n\text{-}1)\times(n\text{-}1)$ Fick diffusion coefficient matrix. For ternary electrolyte mixtures investigated in this work, two mass modes are accessible by DLS. However, available studies in the literature have often reported a single mass mode when it comes to ternary mixtures with low molecular weight components [27,30,31]. As shown by Bardow [27], the reason for observing only a single mass mode is due to the different scattering strength of the two mass modes. Heller et al. [32] successfully demonstrated that both mass modes of ternary non-electrolyte mixtures containing dissolved gases can be obtained by DLS. In the case of ternary electrolyte mixtures, however, accessing both mass modes has not been demonstrated in the literature so far.

The experiments in this work are combined with equilibrium molecular dynamics (MD) simulations. MD simulations enable the determination of various thermophysical properties, including diffusion coefficients, in a predictive manner [33–35]. Furthermore, MD simulations allow to investigate how intermolecular interactions and the fluid structure influence the diffusive mass transport, which helps to establish structure-property relationships. For example, Kiyobayashi et al. [9] used MD simulations to explore how fluid structure influences ion conductivities for ternary electrolyte mixtures containing lithium hexafluorophosphate ([Li][PF_6]) dissolved in EC and DMC, while Borodin et al. [36] focused extensively on fluid structure in the latter system in their Born–Oppenheimer MD simulations and incorporated the findings to examine its influence on self-diffusion coefficients. Previous publications from this working group [37–39] combined DLS with MD simulations and established structure–property relationships for the Fick diffusion coefficients in binary electrolyte mixtures consisting of lithium bis(trifluoromethylsulfonyl)imide ([Li][NTf_2]) dissolved in variety of molecular and ionic liquid solvents. Based on such relationships, simple predictive engineering models can also be developed [40]. While MD simulations have been applied to study the diffusive mass transport in ternary non-electrolyte mixtures [41–43], their use for determining Fick diffusion coefficients in ternary electrolyte mixtures remains limited. In a recent study by Kjelstrup et al. [44], MD simulations were used to determine the elements of the Fick diffusion coefficient matrix for ternary electrolyte mixtures containing a single electrolyte component in the solvent-velocity reference frame. In previous studies, we have demonstrated simulation methodologies for studying Fick diffusion coefficients of binary electrolyte mixtures [37–39]. Within this study, we extend these investigations by establishing consistent simulation protocols for determining Fick diffusion coefficients of ternary electrolyte mixtures containing a single salt component in the amount-averaged reference frame. The transformation of the Fick diffusion coefficients from this reference frame to other reference frames can be achieved via the transformation matrices presented by Klein et al. [20].

The present work combines DLS experiments with MD simulations to improve understanding of diffusive mass transport driven by applied concentration gradients in ternary electrolyte mixtures at temperatures between (293.15 and 373.15) K over a wide range of solvent and solute amount fractions. For this, ternary electrolyte mixtures consisting of the solute [Li][NTf_2] dissolved in binary mixtures of ACN with either DMSO or DMC are investigated. The results from DLS for the eigenvalues of $\boldsymbol{D}$ were used to validate the simulations. Since the eigenvalues are independent of the reference frames, the results from DLS and MD simulation could be compared directly. In the following, first an overview on the theoretical background of multicomponent mass transport in electrolyte mixtures, with a particular focus on the ternary electrolyte mixtures, is given. Thereafter, the investigated samples and the two applied methods DLS and MD simulations are introduced. In the Results and Discussion section, the agreement between the results obtained by DLS and MD simulations is discussed first. Afterward, the results from both methods are used to investigate the influence of the temperature T, mixture composition, and the intermolecular interactions on the diffusive mass transport in ternary electrolyte mixtures.

2 Theoretical Background

There are two primary formalisms to describe diffusive mass transport in multi-component mixtures in the literature: generalized Fick's law [45] and the theory of Maxwell-Stefan (MS)

[46–48]. In the following, a general description of both, particularly in relation to ternary electrolyte mixtures that contain a single electrolyte component, is given. In line with the objectives of the present work, only the pure diffusion in the presence of spatial gradients in concentrations or chemical potentials is considered and the additional effects such as advection, thermodiffusion, or migration of ions in the presence of an external electric field are excluded.

Generalized Fick's law. For an *n*-component mixture, the generalized Fick's law relates the diffusive amount flux of component *i* J_i to a linear combination of the spatial gradients in concentration and can be expressed in the volume-averaged reference frame as [25]

$$J_i^{\mathrm{V}} = -\sum_{j=1}^{n-1} D_{ij}^{\mathrm{V}} \nabla c_j, \quad j = 1, 2, \cdots, n-1. \tag{1}$$

In Eq. (1), c_i represents the concentration of component *i* and D_{ij}^{V} are the Fick diffusion coefficients in this reference frame. With the help of Eq. (1), the diffusive fluxes of all *n* components can be expressed. Due to the closing condition, which couples the fluxes of the individual components, only $(n-1)$ independent diffusive fluxes and concentration gradients can be defined. Accordingly, a square matrix of Fick diffusion coefficients with a dimension of $(n\text{-}1)\times(n\text{-}1)$ is required for characterizing the diffusive mass transport in a general *n*-component mixture. The Fick diffusion coefficients matrix is not symmetric and its entries depend on the reference frame and component ordering [20]. While D_{ii}^{V} are the main Fick diffusion coefficients, describing the diffusive amount flux of component *i* due to its own concentration gradient, D_{ij}^{V} with $i \neq j$ are the off-diagonal or cross Fick diffusion coefficients, accounting for the diffusive amount flux of component *i* due to a concentration gradient of component *j*.

Here, it is important to emphasize that the diffusive fluxes always depend on the reference frame. In the aforementioned case, the fluxes are given based on gradients in *c* and the corresponding reference frame is named the volume-averaged reference frame, as the velocity of the reference frame is calculated on the basis of the partial volumes of the components [20]. Further commonly used reference frames are the mass-averaged and amount-averaged reference frames [20]. In the case of multi-component mixtures, the Fick diffusion coefficients depend on the reference frames and must be labelled accordingly. An exception is given by binary electrolytes and non-electrolyte mixtures, where it could be shown that the Fick diffusion coefficient is independent of the reference frame [20].

In the following, the general description of diffusion in an *n*-component mixture will be applied to the systems investigated in this work, namely ternary electrolyte mixtures. As the name suggests, such mixtures are comprised of three components. The components are an electrolyte, i.e., salt, and two non-electrolyte components, i.e., the solvents. Within this work, the electrolyte component is denoted as component 1 and the non-electrolyte components have the indices 2 and 3. In solution, the electrolyte component 1 dissociates according to [16]

$$\mathit{Component\ 1} \rightarrow \upsilon_{1+}\left[\mathit{Cation\ (1+)}\right]^{z_{1+}} + \upsilon_{1-}\left[\mathit{anion\ (1-)}\right]^{z_{1-}}, \tag{2}$$

where υ_{1+} and υ_{1-} are stochiometric coefficients of the cation and anion, and z_{1+} and z_{1-} are their charge numbers. Because of this, such a ternary electrolyte mixture consists of 4 different species, namely the cations (1+), anion (1–), solvent 2, and solvent 3. For a non-electrolyte mixture consisting of four different species, a Fick diffusion coefficient matrix with the dimension 3×3 is required for describing diffusive mass transport, according to Eq. (1) [49]. However, for a ternary electrolyte mixture, which contains four different species, a 2×2 matrix of Fick diffusion coefficient matrix is required [20,50]. The reason for this is that the movement of the ions in the mixture is coupled due to strong electrostatic forces [25], maintaining macroscopic electroneutrality in the bulk [16]. The electroneutrality constraint demands the same amount of positive and negative charges in a unit volume in the bulk, meaning $\upsilon_{1+}\cdot c_{1+} + \upsilon_{1-}\cdot c_{1-} = 0$ for the present case [10]. Because of this, there are only two independent diffusive fluxes [19] and two independent concentration or chemical potential gradients [46], under the condition of the pure diffusion problem. As a result, Eq. (1) can be expressed for the ternary electrolyte mixture at hand as [20]

$$\begin{pmatrix} J_1^{\mathrm{V}} \\ J_2^{\mathrm{V}} \end{pmatrix} = -\begin{pmatrix} D_{11}^{\mathrm{V}} & D_{12}^{\mathrm{V}} \\ D_{21}^{\mathrm{V}} & D_{22}^{\mathrm{V}} \end{pmatrix} \begin{pmatrix} \nabla c_1 \\ \nabla c_2 \end{pmatrix}. \tag{3}$$

In Eq. (3), c_i and J_i^{V} represent the concentration and the diffusive amount flux of component i, respectively. Therein, component 3 is considered as the dependent component. It is important to note that c_1, i.e., the concentration of the electrolyte component, is related to the concentration of cations and anions according to $c_1 = c_{1+}\cdot\upsilon_{1+}^{-1} = c_{1-}\cdot\upsilon_{1-}^{-1}$ [16,17]. Furthermore, J_1^{V} is related to that of cations J_{1+}^{V} and anions J_{1-}^{V} as $J_1^{\mathrm{V}} = J_{1+}^{\mathrm{V}}\cdot\upsilon_{1+}^{-1} = J_{1-}^{\mathrm{V}}\cdot\upsilon_{1-}^{-1}$. The closing condition, coupling the fluxes of all three components, is given as $J_1^{\mathrm{V}}\overline{V}_1 + J_2^{\mathrm{V}}\overline{V}_2 + J_3^{\mathrm{V}}\overline{V}_3 = 0$ [20], where $\overline{V}_i$ represents the partial molar volume of the component i. Here, the diffusion coefficients and fluxes are given in the volume-averaged reference frame. The diffusion coefficient matrix $\boldsymbol{D}^{\mathrm{V}}$ can be converted into other reference frames, such as the amount-averaged reference frame $\boldsymbol{D}^{\mathrm{M}}$, via the relationships [20]

$$\boldsymbol{D}^{\mathrm{V}} = \left[\boldsymbol{M}\boldsymbol{X}\boldsymbol{\Phi}^{-1}\right]^{-1} \boldsymbol{D}^{\mathrm{M}} \left[\boldsymbol{M}\boldsymbol{X}\boldsymbol{\Phi}^{-1}\right], \tag{4}$$

$$\boldsymbol{\Phi} = \begin{pmatrix} x_1(1-\phi_1) & -x_1\phi_2 \\ -x_2\phi_1 & x_2(1-\phi_2) \end{pmatrix}, \ \boldsymbol{X} = \begin{pmatrix} x_{1*}(1-x_{1*}) & -x_{1*}x_{2*} \\ -x_{1*}x_{2*} & x_{2*}(1-x_{2*}) \end{pmatrix}, \text{ and } \boldsymbol{M} = \begin{pmatrix} 1/\upsilon_1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{5}$$

The matrices in Eq. (5) contain the amount fractions x_i and volume fractions ϕ_i of the components. υ_1 is defined as $\upsilon_1 = \upsilon_{1+} + \upsilon_{1-}$, where υ_{1+} and υ_{1-} are the stoichiometric coefficients of cations and anions due to the dissociation of one salt molecule. x_{i*} represents stoichiometrically weighted amount fractions, defined as $x_{1*} = \upsilon_1 x_1$ for the electrolyte component 1 and $x_{i*} = x_i$ for a non-electrolyte component i. In this way, it is possible to keep the general structure of the transformation matrices similar to those for a non-electrolyte ternary mixture [51]. Similarly, the conversion between matrix $\boldsymbol{D}^{\mathrm{V}}$ and Fick diffusion coefficient matrix in mass-averaged reference frame $\boldsymbol{D}^{\mathrm{m}}$ is possible, as shown by Klein et al. [20].

For the analysis of thermodynamic stability criterion, the analysis of the eigenvalues of the Fick diffusion coefficient matrices is considered [52]. For a ternary mixture in the volume-averaged reference frame, the eigenvalues are

$$\hat{D}_1 = \frac{D_{11}^{\mathrm{V}} + D_{22}^{\mathrm{V}} + \sqrt{\left(D_{11}^{\mathrm{V}} - D_{22}^{\mathrm{V}}\right)^2 + 4D_{12}^{\mathrm{V}} \cdot D_{21}^{\mathrm{V}}}}{2} \text{ and } \hat{D}_2 = \frac{D_{11}^{\mathrm{V}} + D_{22}^{\mathrm{V}} - \sqrt{\left(D_{11}^{\mathrm{V}} - D_{22}^{\mathrm{V}}\right)^2 + 4D_{12}^{\mathrm{V}} \cdot D_{21}^{\mathrm{V}}}}{2}. \tag{6}$$

For a thermodynamically stable fluid, the eigenvalues of the Fick diffusion coefficients matrix must be real and positive in accordance with irreversible thermodynamics [46,52]. Applications of the eigenvalues for analyzing fluxes in multicomponent mixtures are given by Cullinan [53] and documented by Taylor and Krishna [46]. In most cases, the transformations of Fick diffusion coefficients from one reference frame to another are similarity transformations [46]. As a result, the eigenvalues of Fick diffusion coefficient matrices, specifically the in volume-averaged $\boldsymbol{D}^{\mathrm{V}}$, amount-averaged $\boldsymbol{D}^{\mathrm{M}}$, and mass-averaged $\boldsymbol{D}^{\mathrm{m}}$ reference frames, are identical [46]. Furthermore, it is important to note that when one of off-diagonal elements of the Fick diffusion coefficients matrix becomes zero, which results in a triangular matrix, the diagonal elements are equal to the eigenvalues. In such cases, the flux of the corresponding component is governed solely by the associated main diffusion coefficient and only depends on its own concentration gradient.

Maxwell-Stefan approach for a ternary electrolyte system. The MS theory is another robust framework for describing multicomponent diffusive mass transport [46]. This methodology links the driving force for diffusion, i.e., the gradient of the (electro)chemical potential $\nabla \mu_i$ under conditions of constant T and pressure p, to the frictional forces between the molecules of different species in the mixture. For a k-species mixture, the diffusive mass transport is given by [43,46]

$$-\frac{1}{RT}\nabla_{T,p}\mu_i = \sum_{j=1,\, j\neq i}^{k} \frac{x_j\left(u_i - u_j\right)}{Đ_{ij}}. \tag{7}$$

Here, x_j is the amount fraction of species j, and R and T are the gas constant and absolute temperature, respectively. The difference between velocities of species i and j is proportional to the frictional forces between them, and the MS diffusion coefficients $Đ_{ij}$ represent inverse friction coefficients [43]. It is important to note that number of components is essentially the number of species for an n-component non-electrolyte mixture. However, for an n-component electrolyte mixture, $k \geq n + 1$ due to dissociation of the electrolyte components into cations and anions. For the present case, where only a single electrolyte component is present, $k = n + 1$. In Eq. (7), μ_i represents the electrochemical potential for an electrolyte species and the chemical potential for a non-electrolyte species [20].

As both MS theory and Fick's law describe the same physical phenomena, the corresponding diffusion coefficients are related to each other. While these relationships have extensively been applied for non-electrolyte mixtures in the literature [41,43,46,49], we recently showed their relationship for an n-component electrolyte mixture containing a single electrolyte component [20]. For this, a conversion of the MS theory from the species-based representation to the component-based representation via the electroneutrality condition and the vanishing current condition for pure diffusion problem was required. The underlying relationship is given in the following for an n-component electrolyte mixture containing a single electrolyte component, which is of relevance for the present study. In the following, we consider the electrolyte component as component 1 where its cation and anion species are represented by 1+ and 1−. Furthermore, the non-electrolyte components, i.e., the solvents, are denoted by 2, 3, …, n. The resulting relationship in the amount-averaged reference frame takes the form [20]

$$\boldsymbol{D}^{\mathrm{M}} = \boldsymbol{\Delta}^{*}\cdot\boldsymbol{\Gamma}, \text{ with } \boldsymbol{\Delta}^{*} = \boldsymbol{M}\cdot\boldsymbol{B}^{*-1}\cdot\boldsymbol{M}^{-1}. \tag{8}$$

Here, $\boldsymbol{D}^{\mathrm{M}}$, $\boldsymbol{M}$, $\boldsymbol{\Delta}$, $\boldsymbol{B}^{*}$, and $\boldsymbol{\Gamma}$ are matrices with dimensions of $(n\text{-}1)\times(n\text{-}1)$. The asterisk symbol is introduced in matrix $\boldsymbol{\Delta}^{*}$ and $\boldsymbol{B}^{*}$ to distinguish them from the corresponding matrices in non-electrolytes or in a species-based representation. $\boldsymbol{M}$ is a diagonal matrix with $M_{11} = \upsilon_1^{-1}$, $M_{ii,i\neq 1} = 1$, and $M_{ij,i\neq j} = 0$. The elements of $\boldsymbol{B}^{*}$ are

$$B^{*}_{ii} = \frac{K_{in}}{x_{n*}} + \sum_{j=1;\, j\neq i}^{n} \frac{K_{ij}}{x_{i*}} \text{ with } i = (1, \ldots, n-1) \text{ and}$$

$$B^{*}_{ij,\, j\neq i} = \frac{K_{in}}{x_{n*}} - \frac{K_{ij}}{x_{j*}} \text{ with } i, j = (1, \ldots, n-1) \text{ and } i \neq j. \tag{9}$$

In Eq. (9), the quantity K_{ij} relates to the MS diffusion coefficients as shown below and is adopted from the work of Newman and coworkers [10,54] for the convenience of the derivations of the set of equations, see the work of Klein et al. [20]. Consequently, the relationship between K_{ij} and the MS diffusion coefficients, which contain the electrolyte component 1 and a non-electrolyte component j, can be expressed as

$$K_{1,j} = K_{j,1} = \frac{x_{1*}x_{j*}}{Đ_{1,j}} = \left(\frac{\upsilon_{1+}/\upsilon_1}{Đ_{1+,j}} + \frac{\upsilon_{1-}/\upsilon_1}{Đ_{1-,j}}\right) x_{1*}x_{j*} \text{ with } j \neq 1. \tag{10}$$

It is important to note that $Đ_{1,j}$ in Eq. (10) represents a component-based MS diffusion coefficient. Furthermore, its relationship to the species-based MS diffusion coefficients, namely the $Đ_{1+,j}$ and $Đ_{1-,j}$, takes essentially the same form, as given by Newman and Thomas-Alyea for a binary electrolyte mixture [10]. The relationship between K_{ij} and the respective MS diffusion coefficient of a pair of two non-electrolyte components is given by [20]

$$K_{i,j} = \frac{x_{i*}x_{j*}}{Đ_{ij}}. \tag{11}$$

The matrix $\boldsymbol{\Gamma}$ in Eq. (8) is the so-called thermodynamic factor matrix and its elements are defined according to

$$\Gamma_{ij} = \delta_{ij} + x_i \left(\frac{\partial \ln \gamma_i}{\partial x_j} \right)_{T,p,\Sigma} , \qquad (12)$$

where δ_{ij} is the Kronecker delta and γ_i represents the mean activity coefficient of the electrolyte component or the activity coefficient of a non-electrolyte component on the amount fraction scale. The symbol Σ represents a constrained derivative, meaning that it is performed while maintaining constant amount fractions of all other species except the n^{th} species, ensuring that the sum of the amount fractions of all species equals unity.

Eq. (8) relates the Fick diffusion coefficient matrix to the MS diffusion coefficients via the thermodynamic factor matrix. While experimental methods can provide access to Fick diffusion coefficients, they do not offer a direct means to determine the MS diffusion coefficients. In contrast, MD simulations offer valuable insights by effectively facilitating the determination of MS diffusion coefficients as well as the elements of the thermodynamic factor matrix, and thus, the Fick diffusion coefficient matrix, with detailed information provided in section 3.2.

3 Methodology

3.1 Materials and Sample Preparation

Within this work, ternary mixtures containing the solute lithium bis(trifluoromethylsulfonyl)imide ([Li][NTf_2]) dissolved in binary mixtures of acetonitrile (ACN) and dimethyl carbonate (DMC) or dimethyl sulfoxide (DMSO) were investigated across various solute and solvent amount fractions over the T range between (293.15 and 373.15) K. In addition, the binary subsystems of solvents were investigated in the present study. The results for the binary subsystems containing [Li][NTf_2] can be found in our previous studies [37,55]. Details about the investigated substances, including information about the suppliers, purity, and molecular masses of each substance, are given in Table 1. In addition, Table 1 also lists further properties of the solvents, namely their liquid viscosity η and static dielectric constants ε_r at T = 298.15 K. In the following, the components in the ternary mixtures are abbreviated as follows. [Li][NTf_2] is designated as component 1. For the solvents combination of ACN and DMC, ACN is assigned as component 2 and DMC as component 3. In the case of ACN and DMSO mixtures, ACN is designated as component 2 and DMSO as component 3.

Table 1. Specification of the Chemical Supplier, Mass or Volume Fraction Purity *w* or ϕ, Molecular Mass *M*, as Well as the Solvent Dynamic Viscosity η and Static Dielectric Constant ε_r at *T* = 298.15 K and Ambient *p*.

component	CAS reg. no.	source	w or ϕ purity	M / (g·mol^{-1})	η / (mPa·s)	ε_r
acetonitrile (ACN)	75-05-8	Sigma Aldrich	$w \geq$ 0.9993	41.05	0.341	35.9
dimethyl sulfoxide (DMSO)	67-68-5	Sigma Aldrich	$w \geq$ 99.9	78.13	1.948	46.4
dimethyl carbonate (DMC)	616-38-6	Sigma Aldrich	$w \geq$ 99	90.08	0.629	3.1
lithium bis(trifluoromethyl-sulfonyl)imide ([Li][NTf_2])	90076-65-6	Solvionic	$w \geq$ 99.9	287.1	-	-
nitrogen (N_2)	7727-37-9	Air Liquide	ϕ = 0.999999	28.01	-	-
helium (He)	7440-59-7	Linde AG	ϕ = 0.999993	4	-	-

[a]Purities of the solvents and solute as specified in the certificate of analysis of the supplier. [b]η and ε_r at T = 298 K are taken from Zhang et al. [56].

Due to the hydrophilic nature of electrolyte mixtures, all the sample preparation steps were conducted inside a glovebox flushed continuously with N_2 at ambient T. All mixtures were prepared using a mass balance with an expanded uncertainty (coverage factor k = 2) of 1 mg. First, the binary solvent mixtures ACN and DMC or ACN and DMSO were prepared at amount fractions of ACN x_2' of approximately (0.25, 0.50, and 0.75). Here, x' is used to differentiate it from the amount fractions of a ternary mixture x. Prior to adding the electrolyte salt, each binary solvent mixture was stirred for at least two hours at ambient T. Afterward, the ternary mixtures were prepared by the addition of desired amount of the solute $[Li][NTf_2]$ into binary mixture of solvents. The amount fraction of each component i x_i in the resulting ternary mixtures is listed in Table 2. To ensure homogeneity, each ternary mixture was subsequently stirred for at least 8 hours at ambient T. To remove particles like impurities, which could be present in the pure components and capable of affecting the DLS measurements, all mixtures were filtered through a stacked set of two hydrophilic PTFE filters with pore sizes of 450 nm and 200 nm. After the DLS experiments, the water content of the samples was measured by Karl Fischer coulometric titration. The determined water contents are listed in Table 2 and represent average values from three readings. The relative statistical uncertainty (k = 2) of the water content is less than 20%. The influence of the water content on mass diffusivities was investigated in one of our prior studies using binary electrolyte mixtures as a model system [37]. Therein, the determined mass diffusivities remained unchanged within experimental uncertainties for water contents up to about 15,000 ppm. These experimental observations indicated that water content within the investigated range has no measurable effect on mass diffusivities. In this connection, we conclude that the water content of the samples in the present study, which is well below 15,000 ppm, has a negligible effect on the mass diffusivities obtained from DLS.

Table 2. Amount of Substance Fraction of the Components in the Ternary Mixtures x or Binary Solvent Mixture x' and Water Mass Fraction w_{water} of the Investigated Ternary Mixtures Determined after Each Experimental Series.[a]

Ternary systems			Binary solvent subsystems		$10^6 \times w_{water}$ [b]
x_1	x_2	x_3	x_2'	x_3'	
$[Li][NTf_2]$ (1) in ACN (2) and DMC (3)					

0.0500	0.2379	0.7121	0.2504	0.7496	2520.43
0.0500	0.4749	0.4751	0.4999	0.5001	3449.17
0.0500	0.7129	0.2371	0.7504	0.2496	2520.43
[Li][NTf_2] (1) in ACN (2) and DMSO (3)					
0.0499	0.2375	0.7126	0.2500	0.7500	5691.00
0.0435	0.4783	0.4782	0.5001	0.4999	3075.40
0.0370	0.7222	0.2408	0.7500	0.2500	3023.57
0.1000	0.4501	0.4499	0.5001	0.4999	2476.77
0.1497	0.4252	0.4250	0.5001	0.4999	3770.53
0.2000	0.4008	0.3993	0.5009	0.4991	3411.03
0.2500	0.3747	0.3753	0.4996	0.5004	2162.15

[a]The estimated absolute expanded uncertainty of the amount of substance fractions is 0.001 (k = 2). [b]The expanded relative uncertainty (k = 2) of w_{water} is less than 20%.

3.2 Dynamic Light Scattering (DLS)

DLS is a versatile experimental technique that provides access to several thermophysical properties of fluids, often simultaneously. For studying diffusive mass transport, DLS has become a standard method for determining mutual diffusion coefficients in both binary and multicomponent mixtures. In binary mixtures, DLS measures the Fick diffusion coefficient of the system [57–60]. In ternary mixtures, it provides the two eigenvalues of the Fick diffusion coefficient matrix [27,32]. It is important to note that the eigenvalues and the elements of the Fick diffusion coefficient matrix are fundamentally different. The eigenvalues coincide with the diagonal elements of the matrix only when one or both cross-Fick diffusion coefficients are absent. A detailed description on the theory and experimental realization of DLS is available in the literature [26,27,34,55,58,59,61,62]. In the following, only the information relevant for the present study is given.

DLS is a contactless experimental technique, which operates at macroscopic thermodynamic equilibrium. Based on rigorous working equations, DLS provides access to various thermophysical properties, including diffusion coefficients, without the need for calibration [59]. During DLS experiments, a fluid sample at macroscopic thermodynamic equilibrium is irradiated by coherent laser light. The scattering process is influenced by spontaneous microscopic fluctuations in T, p, and species concentrations c [26,59]. In other words, the scattered light inherently encodes the information about these fluctuations. By analyzing the dynamics of the scattered light intensity, DLS enables the study of the relaxation of fluctuations of these state variables [26,58,59], which are governed by the same physical laws describing their decay in a macroscopic non-equilibrium system, according to Onsager's regression hypothesis [28,29]. The analysis is usually performed by the calculation of a normalized second-order correlation function (CF) of the scattered light intensity [59]. According to the theory of DLS for a ternary mixture and neglecting the p fluctuations, the CF under the heterodyne detection scheme, where the scattered light is superimposed with a strong coherent local oscillator field, reads [32]

$$g^{(2)}(\tau) = b_0 + b_\mathrm{t}\exp(-|\tau|/\tau_\mathrm{C,t}) + b_\mathrm{c,1}\exp(-|\tau|/\tau_\mathrm{C,c1}) + b_\mathrm{c,2}\exp(-|\tau|/\tau_\mathrm{C,c2})\,. \tag{13}$$

Here, the constants b_0, b_t, $b_\mathrm{c,1}$ and $b_\mathrm{c,2}$ are experimental constants which depend on characteristics of the experimental setup, thermodynamic state, and the strength of the local oscillator field. $\tau_\mathrm{C,t}$ and $\tau_{\mathrm{C,c}i}$ are the mean lifetimes or mean decay times of the fluctuations in T and c, respectively, and reflect their relaxation. They are related to the thermal diffusivity a and the two eigenvalues of the square Fick diffusion matrix $\boldsymbol{D}^\mathrm{V}$, which are abbreviated within this study as $\hat{D}_1$ and $\hat{D}_2$, according to [32]

$$a = \frac{1}{q^2\tau_\mathrm{C,t}},\quad \hat{D}_1 = \frac{1}{q^2\tau_\mathrm{C,c1}},\quad \hat{D}_2 = \frac{1}{q^2\tau_\mathrm{C,c2}} \tag{14}$$

Here, the coupling between hydrodynamic modes related to thermal and mass diffusivities are neglected, which is safely valid for the mixtures studied within this work [32,63]. In Eq. (14), q is the modulus of the scattering vector, which can be calculated according to

$$q = (4\pi n_\mathrm{fluid}/\lambda_0)\sin(\Theta_\mathrm{S}/2)\,, \tag{15}$$

where n_fluid is the refractive index of the fluid and λ_0 the wavelength of the laser in vacuo. The scattering angle, denoted as Θ_S, is determined using Snell−Descartes law and the directly accessible incident angle Θ_I. The latter is defined by the direction of the incident light and that of the detected scattered light.

In some of the thermodynamic states studied in this work, $\tau_{\mathrm{C,c}i}$ was much larger than $\tau_{\mathrm{C,t}}$. This means that the c fluctuations decay more slowly than the T fluctuations. Due to the focus of this work on mass diffusion coefficients, the temporal resolution of the CFs was accordingly adjusted to maximize the resolution of the c fluctuations and minimize the uncertainties of the related eigenvalues. As a result, the information on the faster thermal mode was only contained in the first few data points of the CFs, which were often insufficient to resolve a. Thus, the determination of the a for such systems was not possible. In this context, corresponding data points associated with T fluctuations are neglected during data evaluation, and the rest of the data points are fitted using only two exponential functions in Eq. (13), reflecting the c fluctuations. As a result, Eq. (13) reduces to

$$g^{(2)}(\tau) = b_0 + b_{\mathrm{c},1}\exp(-|\tau|/\tau_{\mathrm{C,c1}}) + b_{\mathrm{c},2}\exp(-|\tau|/\tau_{\mathrm{C,c2}}) . \quad (16)$$

Details on the optical and electronic setups can be found in previous publications [64]. After realizing macroscopic thermodynamic equilibrium, four to six consecutive experimental runs were performed for each thermodynamic state at different q values. Here, most thermodynamic states were investigated at lower q values, corresponding to smaller Θ_S values. This is motivated by the improved statistical quality of the measured CFs with short measurement times under such conditions. In detail, because the wavelength or size of underlying fluctuations scales inversely with q [65–67], smaller q values allow the study of c fluctuations at larger length scales. These fluctuations with larger length scales usually produce higher scattering intensities. This improves signal statistics, enables faster data acquisition, and allows diffusion coefficients to be determined with lower uncertainty. Here, each experimental run lasted about 20 minutes. An exception to this was the mixture containing 0.25 $[\mathrm{Li}][\mathrm{NTf_2}]$ dissolved in an equimolar binary mixture of ACN and DMSO at T

= 293.15 K. Owing to the higher viscous nature and consequently sluggish dynamics of this mixture at the specified T, $\tau_{C,ci}$ of the c fluctuations studied at small q values occurred over a timescale of thousands of milliseconds. This long timescale is highly susceptible to external effects, such as vibrations from laboratory equipment, which complicate the data evaluation and compromise the reliable determination of diffusion coefficients. To obtain reliable diffusivity data for such systems with slow dynamics, it is therefore advantageous to probe c fluctuations with smaller length scales and correspondingly shorter relaxation times, where such a time scale is not affected by external perturbations. This can be achieved by DLS investigations at larger q values, probing c fluctuations with shorter length and time scales. For this reason, the mixture with $x_{[Li][NTf2]}$ = 0.25 at the lowest investigated T was studied at higher q values at Θ_I around 90°. However, investigations at larger q generally yield lower scattering intensities, consequently, reducing the statistical quality of the CFs. To obtain statistically sound enough CFs for a reliable data evaluation, longer statistical averaging is, therefore, needed. This resulted in an extended experimental data acquisition period of about 12 hours for this specific state point. Although experimentally demanding, this trade-off is necessary to ensure robust data analysis.

For the determination of corresponding diffusivities in accordance with Eq. (14), knowledge of the q and associated mean lifetime of the fluctuations is necessary. The determination of q relies on the value of λ_0, n_{fluid} and Θ_S. The λ_0 in the present study is 532 nm. Snell−Descartes law enables the determination of Θ_S using Θ_I, n_{fluid} and refractive index of air n_{air}. For the investigations at smaller q, where Θ_I was varied between –10° and +9°, the calculation of q is based on $q = 4 \cdot \pi \cdot n_{fluid} \cdot \lambda_0^{-1} \cdot \sin\left[0.5 \cdot \arcsin\left(\sin\left(\Theta_I \cdot n_{air} \cdot n_{fluid}^{-1}\right)\right)\right]$. Since, for a given angle Θ with a small value, $\sin(\Theta) \approx \Theta$, and n_{air} has a value of approximately unity, a change in n_{fluid} has a negligible influence on the determined q and thus the diffusivities [68]. For example, even a change of n_{fluid} by 300% would result in a change of the diffusivities about 0.61%, which is clearly within the uncertainty of the measured diffusivities. Thus, an estimated value of n_{fluid}, based on refractive indices of solvents, is used for the determination of the q and diffusion coefficients under such small angles of Θ_{I}. Again, an exception here is the mixture with $x_{\text{[Li][NTf2]}} = 0.25$ at the lowest investigated T. As mentioned above, this mixture was studied at larger scattering or incident angles around 90°. For such large incident angles, accurate values for n_{fluid} are, however, required for a reliable determination of q and, subsequently, the diffusivities. Thus, n_{fluid} for the this mixture was measured at different temperatures using an Abbe refractometer, for which the details are given in a previous publication [55]. Afterward, n_{fluid} at the thermodynamic states investigated by DLS could be obtained from T-dependent correlations of the measured values, which are reported in Table S1 in the Supporting Information.

For the determination of the CFs, and subsequently the mean lifetimes of the fluctuations in accordance with Eqs. (13 or 16), the scattered light was detected using photo multiplier tubes (PMTs). To suppress artefacts such as deadtime and after-pulsing effects, the two PMTs were configured in a pseudo cross-correlation scheme. The CFs were then computed by two digital correlators, namely a linear-tau correlator with 2047 equally spaced channels and a multi-tau correlator with a logarithmic time structure featuring 263 channels. The mean lifetimes of the fluctuations as well as the experimental constants were determined through least-squares fitting of the recorded CFs to Eq. (13 or 16). In addition to the modes related to thermal and mass diffusivities, the CFs may contain further contributions, which can arise from the stray light in the laboratory, dust on the mirrors, and/or particle-like impurities in the samples. To account for these effects, additional first- and/or second-order polynomials are added to Eq. (13 or 16). The reported $\hat{D}_i$ values at each thermodynamic state represent weighted averages of 8–12 CFs, obtained from four to six independent experiments.

The reported uncertainties for each thermodynamic state are expanded experimental uncertainties (k = 2).

Diffusivities accessible by DLS experiments and proof of hydrodynamic modes. According to the theory of DLS for ternary mixtures, three modes should be accessible by DLS [27,32]. One is the thermal mode, which is associated with thermal diffusivity. The other two are the mass modes, which are related to the two eigenvalues of the square Fick diffusion coefficient matrix [27,32]. However, for the ternary mixtures containing a low amount fraction of [Li][NTf_2], only one mass mode could be assessed within this study. Similar observations have been reported in the literature for ternary mixtures with molecular fluids [31,62]. The reason behind this could be explained by Bardow [27]. By studying ternary mixtures consisting of glycerol, acetone, and water, he demonstrated that the observation of only a single mass mode is due to the different scattering intensities of the different modes. However, he also pointed out the potential of resolving both mass modes during DLS experiments, depending on the mixture compositions. The mixtures for which two mass modes could be accessed within this work are presented later in this section. Figure 1 illustrates an example of a CF for which only one mass mode, next to the thermal mode, can be resolved. It was recorded for the mixture containing 0.05 [Li][NTf_2] dissolved in a binary solvent mixture of 0.25 ACN and 0.75 DMSO at T = 372.97 K. The residual plot in the lower part of Figure 1 shows that the fit model in Eq. (13) accurately represents the experimental signal.

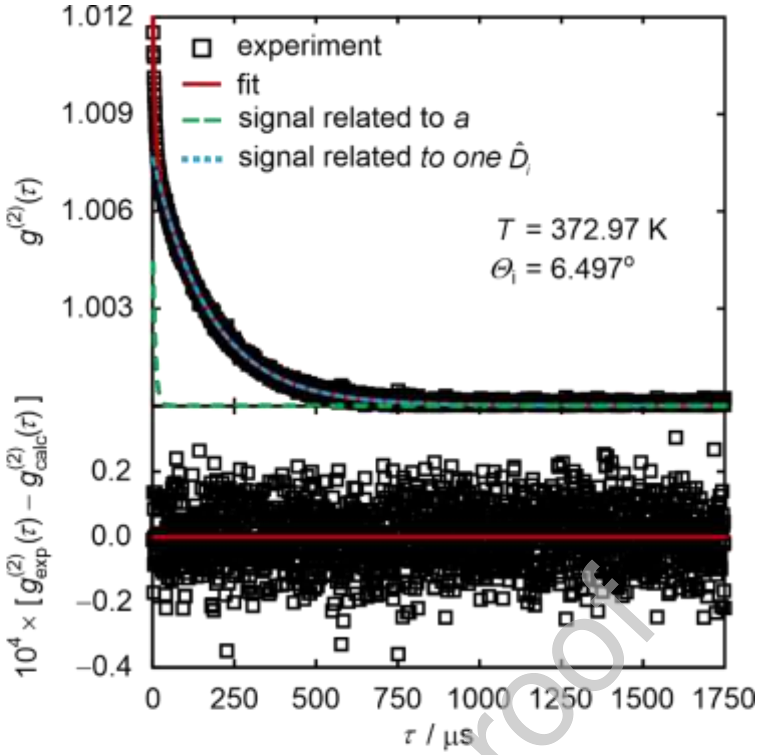

Figure 1. Correlation function (CF) (top) and the residuals of the measured CF from the fit (bottom) for the ternary mixture consisting of [Li][NTf_2] dissolved in a binary mixture of solvents 0.25 ACN + 0.75 DMSO with $x_{[Li][NTf2]}$ = 0.05 at T = 372.97 K.

To clarify whether the resolved modes originate from hydrodynamic fluctuations, it is essential to analyze the dependency of mean lifetime of their fluctuations on q^2. According to the theory of hydrodynamic fluctuations [69], this relationship should be linear and pass through the origin. Figure 2 exemplifies this for the same mixture discussed previously. The solid lines represent a linear fit of the inverse of the mean lifetime as a function of q^2, weighted by the inverse of the uncertainties. The fits were not forced to pass through the origin. Taking the measurement uncertainties into account, the linear fits cross the origin, which confirms that the resolved modes are indeed related to hydrodynamic fluctuations. The discussion of whether the resolved mass mode is the faster or slower one will be addressed in the Results and Discussion section.

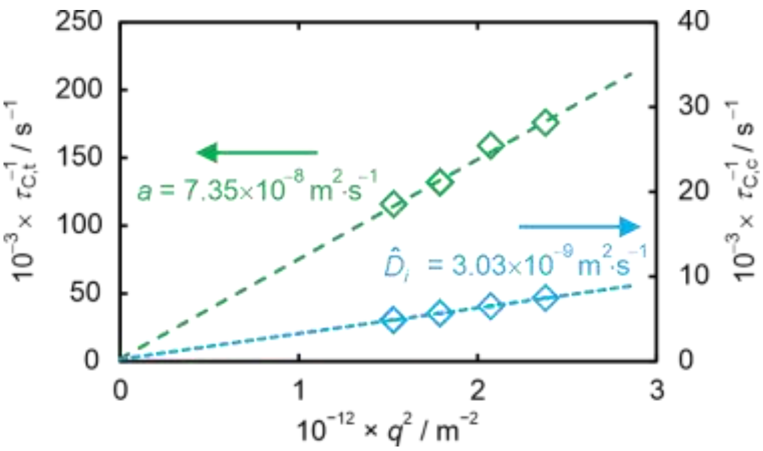

Figure 2. Inverse of the mean lifetimes related to thermal diffusivity $\tau_{C,t}^{-1}$ or to one of the mass modes $\tau_{C,c}^{-1}$, as a function of the squared modulus of the scattering vector q^2 for the ternary mixture consisting of [Li][NTf_2] dissolved in a binary mixture of solvents 0.25 ACN + 0.75 DMSO with $x_{[Li][NTf2]} = 0.05$ at $T = 372.97$ K.

For the ternary mixtures containing high amounts of [Li][NTf_2], particularly those with $x_{[Li][NTf2]}$ greater than 0.15, it was possible to resolve both mass modes. An example of a CF is shown in Figure 3 for the mixture containing [Li][NTf_2] dissolved in an equimolar binary mixture of ACN and DMSO with $x_{[Li][NTf2]} = 0.20$ at $T = 347.35$ K. Here, the temporal resolution of the CF by the linear-tau correlator is adjusted to maximize the information relevant for the mass modes. As a result, the faster thermal mode could not be resolved in this case and is not depicted in Figure 3. Consequently, the fitting model is provided by Eq. (16). The residual plot at the bottom part of Figure 3 confirms that the fit model accurately represents the experimental signal.

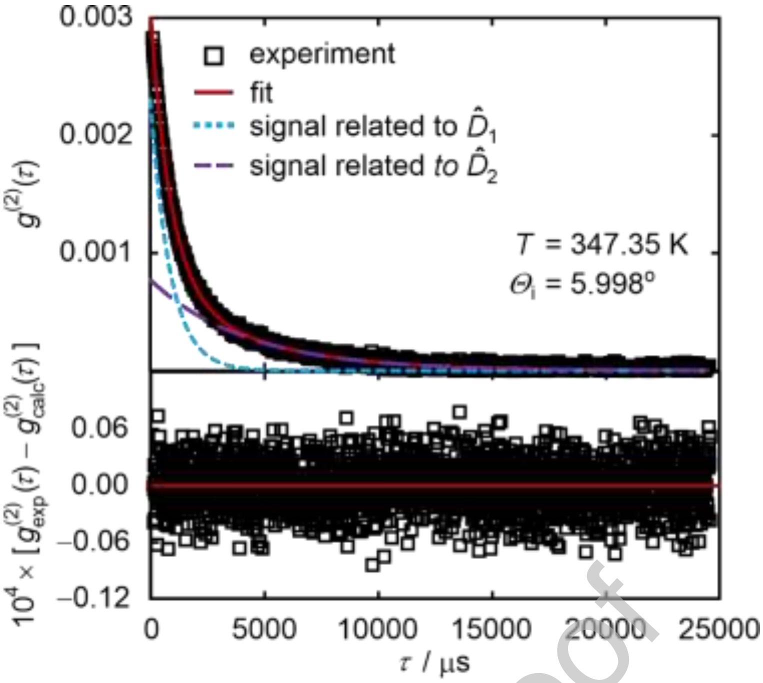

Figure 3. Correlation function (CF) (top) and the residuals of the measured CF from the fit (bottom) for the ternary mixture consisting of [Li][NTf_2] dissolved in an equimolar mixture of ACN and DMSO with $x_{[Li][NTf2]} = 0.20$ at $T = 347.35$ K.

Also for this system, the dependency of the mean lifetime of the fluctuations on the q^2 was studied, which is depicted in Figure 4, exemplary for the mixture containing [Li][NTf_2] dissolved in an equimolar binary solvent mixture of ACN and DMSO with $x_{[Li][NTf2]} = 0.20$ at $T = 322.81$ K. Even though the thermal mode is not visible in linear-tau CF in Figure 3, it could be accessed from the CF recorded with the multi-tau correlator, which operates in parallel to the linear-tau correlator. Therefore, the q^2 dependency for the thermal mode as resolved using the multi-tau correlator is also presented in Figure 4. The solid lines represent linear fits of the inverse of the mean lifetimes as a function of q^2, weighted by the inverse of the uncertainties. The fits were not forced to pass through the origin. Considering the experimental uncertainties, the linear fits for three modes intersect at the origin, confirming that the resolved modes are related to hydrodynamic fluctuations.

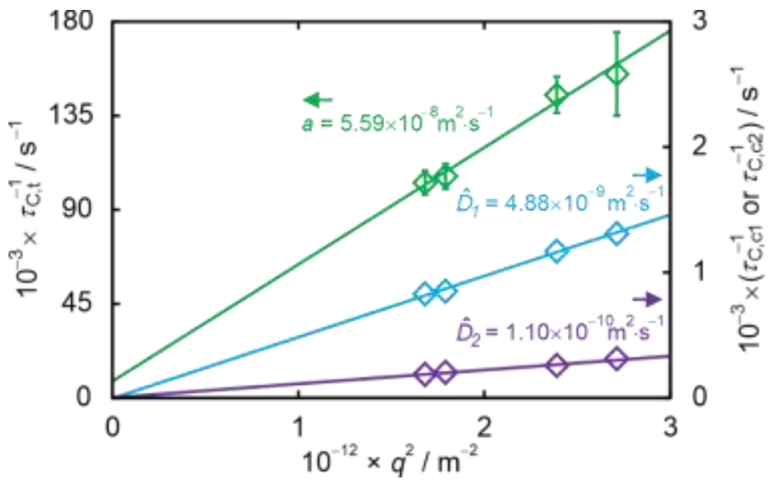

Figure 4. Inverse of the mean lifetimes $\tau_{C,t}$, $\tau_{C,c1}$, and $\tau_{C,c2}$ as a function of the squared modulus of the scattering vector q^2 for the ternary mixture containing [Li][NTf$_2$] dissolved in an equimolar binary solvent mixture of ACN and DMSO with $x_{[Li][NTf2]} = 0.20$ at T = 322.81 K. Error bars smaller than the symbol sizes are not shown.

3.3 Molecular Dynamics (MD) Simulations

MD simulations solve Newtonian classical equations of motion in discretized time steps and allow for the prediction of the movement of molecules and atoms in a fluid ensemble [70–72]. The resulting trajectories and energy files can be analyzed to access various thermophysical properties, including diffusion coefficients [34,43,73]. The reliability of the predicted properties mainly relies on the so-called force field (FF). The FF defines how two interaction sites interact with each other, whether they are part of different molecules or within the same molecule [70–72]. Additional information on the basic principles of MD simulations can be found in the literature [70–72,74]. The following discussion focuses solely on aspects pertinent to the mixtures examined in this study.

Simulation details. The MD simulations were carried out using the GROMACS software package version 2021.5 [75] and executed on high-performance computing nodes equipped with AMD Milan CPUs and Nvidia A40 GPUs. Interactions within and between molecules were modeled using pair-additive, non-polarizable FFs following the all-atom approach on the basis of the Optimized Potentials for Liquid Simulations (OPLS) [76]. The leapfrog algorithm was employed to integrate the equations of motion. For the calculation of the Lennard-Jones (LJ) interactions between unlike atom types, the geometric combination rule was used [76]. A cutoff radius of 1.2 nm was set for both electrostatic and LJ interactions. Beyond this range, long-range LJ contributions were corrected using standard dispersion corrections [77], while electrostatic interactions were computed using the particle mesh Ewald (PME) summation [78]. A scaling factor of 0.5 was applied to 1-4 intramolecular interactions, referring to two atoms separated by three covalent bonds. All bond lengths were constrained using the linear constraint solver (LINCS) [79]. T and p in the corresponding ensembles were adjusted using the v-rescale thermostat [80] with a coupling time of 1 ps and the Parrinello-Rahman barostat [81,82] with a coupling time of 7 ps, respectively.

The simulations were initiated by generating initial configurations using the PACKMOL software [83]. Initially, solvent molecules were placed in the simulation box, which had periodic boundary conditions applied in all three directions, followed by the addition of the solute ions. The simulation boxes were created with approximately 40,000 interaction sites. This number of interaction sites is selected in connection with addressing finite size effects on diffusivities, for which the detail for the present study are provided in the Supporting Information. The number of molecules used for the simulations is given in Table S2 in the Supporting Information. Following an energy minimization, the systems were equilibrated at the desired T and p in an isothermal-isobaric (NpT) ensemble for 2 ns. The pressure during the NpT simulation was set 0.2 MPa above the vapor pressure of the higher volatile solvent in each system. The thermophysical properties of interest were calculated from subsequent simulations, conducted in the canonical ensemble (NVT) with a 2 fs time step. The total simulation time was approximately 200 ns.

Force fields. The force fields (FFs) for the solvents, namely, DMSO and DMC, are identical to the ones in our previous studies [38]. Therein, the FF parameters were obtained from the OPLS FF and reparametrized. This reparameterization was essential to incorporate polarization effects within the non-polarizable FF [37–39], and thereby enhancing the reliability of predictions for dynamic properties, including diffusivities. In comparison to our previous studies, a new FF for ACN, documented in a publication by the Jorgenson group [84], was utilized within this study. However, to achieve this, the partial charges of this FF needed to be adjusted to reproducing static and dynamic properties of pure ACN. Additional details regarding these adjustments can be found in the Supporting Information. Consistent with our previous work [37–39], the FF developed by Soetens et al. [85] was employed for the $[Li]^+$ cation. The FF used for $[NTf_2]^{-1}$ is the same as the one used in our previous studies [34,37–39].

Data evaluation. Self-diffusion coefficients D_i are determined by examining the mean squared displacement of the species' center of mass in relation to the correlation time τ, as described by Einstein, according to [86]

$$D_i = \frac{1}{6N_i} \lim_{m\to\infty} \frac{1}{m.\Delta\tau} \left\langle \left(\sum_{l=1}^{N_i} \left(r_{l,i}(\tau + m.\Delta\tau) - r_{l,i}(\tau) \right)^2 \right) \right\rangle . \tag{17}$$

Here, the angle brackets represent the ensemble average. N_i is the total number of molecules of species i in the simulation box and m denotes the number of correlation time steps $\Delta\tau$. The term $r_{l,i}(\tau)$ indicates the position of the l^{th} molecule of type i at τ.

Within this study, the matrix of the Fick diffusion coefficients of ternary electrolyte mixtures is determined by the separate evaluation of the elements of $\boldsymbol{B}^*$ and $\boldsymbol{\Gamma}$, according to Eqs. (8-12). For the determination of the component-based MS diffusion coefficients, species-based $Đ_{ij}$ are required. Within this study, $Đ_{ij}$ were calculated via the so-called Onsager coefficients Λ_{ij}, which are directly accessible by MD simulations [35]. Using the relationships given by Krishna and Van Baten [87], approaches for accessing $Đ_{ij}$ via Λ_{ij} for electrolyte mixtures containing three species have been reported by Liu et al. [33]. These relationships for the four species are also documented in the work of Krishna and Van Baten [87]. For this, a matrix $\boldsymbol{\Delta}$ with dimension 3×3 is defined, where its elements are related to Λ_{ij} according to [87]

$$\Delta_{ij} = (1 - x_i)\left(\frac{\Lambda_{ij}}{x_j} - \frac{\Lambda_{i4}}{x_4}\right) - x_i \sum_{k \neq i = 1}^{4} \left(\frac{\Lambda_{kj}}{x_j} - \frac{\Lambda_{k4}}{x_4}\right). \tag{18}$$

Here, x_i represents the amount fraction of each species in the mixture. The elements of the inverted matrix $\boldsymbol{\Delta}^{-1} = \boldsymbol{B}$ relate the MS diffusivities to the elements of $\boldsymbol{B}$ according to [87]

$$Đ_{14} = \frac{1}{B_{11} + \frac{x_2}{x_1} B_{12} + \frac{x_3}{x_1} B_{13}},\ Đ_{24} = \frac{1}{B_{22} + \frac{x_1}{x_2} B_{21} + \frac{x_3}{x_2} B_{23}},\ Đ_{34} = \frac{1}{B_{33} + \frac{x_1}{x_3} B_{31} + \frac{x_2}{x_3} B_{32}},$$

$$Đ_{12} = \frac{1}{\frac{1}{Đ_{24}} - \frac{B_{21}}{x_2}},\ Đ_{13} = \frac{1}{\frac{1}{Đ_{14}} - \frac{B_{13}}{x_1}},\ \text{and } Đ_{23} = \frac{1}{\frac{1}{Đ_{24}} - \frac{B_{23}}{x_2}}. \tag{19}$$

The Onsager coefficients Λ_{ij} are directly computed by MD simulations according to [35]

$$\Lambda_{ij} = \frac{1}{6} \lim_{m \to \infty} \frac{1}{N} \frac{1}{m \cdot \Delta\tau} \times \left\langle \left(\sum_{l=1}^{N_i} \left[r_{l,i}(\tau + m \cdot \Delta\tau) - r_{l,i}(\tau) \right] \right) \left(\sum_{k=1}^{N_k} r_{k,i}(\tau + m \cdot \Delta\tau) - r_{k,i}(\tau) \right) \right\rangle. \tag{20}$$

Here, N represents the total number of molecules in the simulation box, while i and j denote the types of the species. The position of the l^{th} molecule of species i at correlation time τ is given by $r_{l,i}(\tau)$.

The elements of the thermodynamic factor matrix can be determined using microscopic structural information obtained from radial distribution functions (RDFs) $g_{i\square}(r)$, in accordance with the Kirkwood–Buff theory combined with the indistinguishable ion approach. Within the framework of the Kirkwood–Buff theory, the Kirkwood–Buff integrals (KBIs) $G_{i\square}$ are defined in the grand canonical (μVT) ensemble [88,89]. However, MD simulations in an open ensemble are difficult to realize in the condensed liquid phase [90], particularly for electrolyte mixtures due to the electroneutrality condition [88]. Methods have been developed in the past for the calculation of G_{ij} from MD simulations in other ensembles, such as the *NVT* ensemble [91–93]. In detail, G_{ij} can be determined by extrapolating KBIs of finite sub-volumes embedded in closed systems to the thermodynamic limit [91]. Several approaches for performing this extrapolation have been summarized by Dawas et al. [93]. Within this study, G_{i}□ are obtained by applying the expression of the KBI for finite volumes and extrapolating the obtained values to the thermodynamic limit [91]. The application of this approach for electrolyte mixtures was documented by Schnell et al. [88]. Additionally, the RDFs from MD simulations of closed and finite systems are subjected to finite-size effects, which must be corrected. Various methods are available in the literature to address these effects [91,94,95]. In this study, the RDFs are corrected for finite-size effects according to the approach outlined by Ganguly and van der Vegt [95].

However, the use of KBIs for the calculations of the activity derivatives in electrolyte mixtures comes with further complications as the electrolyte components are dissociating into cations and anions, as highlighted by Ben-Naim [89]. As an alternative, the indistinguishable ion approach has been established in the literature to mitigate these issues and obtain thermodynamic properties of electrolyte mixtures [96–98]. In this approach, the cations 1+ and anions 1− are treated as indistinguishable and considered as a pseudo pure component, abbreviated as component *c* here onwards. The KBIs related to the electrolyte species are related to those of the indistinguishable electrolyte component with itself G_{cc} or with another non-electrolyte component *j*, denoted as G_{cj}, according to [98]

$$G_{\mathrm{cc}} = \left(\frac{\upsilon_{1+}}{\upsilon_1}\right)^2 G_{1+,1+} + \left(\frac{\upsilon_{1-}}{\upsilon_1}\right)^2 G_{1-,1-} + \frac{2\upsilon_{1+}\upsilon_{1-}}{\upsilon_1^2} G_{1+,1-} \text{ and } G_{\mathrm{c}j} = \left(\frac{\upsilon_{1+}}{\upsilon_1}\right) G_{1+,j} + \left(\frac{\upsilon_{1-}}{\upsilon_1}\right) G_{1-,j}. \qquad (21)$$

The resulting six KBIs are, namely, G_{cc}, G_{c2}, G_{c3}, G_{22}, G_{23} and G_{33}, where subscripts 2 and 3 refer to non-electrolyte solvents. With the help of G_{ij}, the thermodynamic factors for the pseudo ternary mixture can be calculated using the expressions given by Liu et al. [43]. They can be named as Γ_{cc}, Γ_{c2}, Γ_{2c} and Γ_{22}, which are related to the elements of the thermodynamic factor matrix defined by Eq. (12) via the Gibbs–Duhem relationship according to

$$\Gamma_{11} = \Gamma_{CC}\,,\ \Gamma_{12} = \frac{1}{\upsilon_1}\Gamma_{C2}\,,\ \text{and}\ \Gamma_{21} = \upsilon_1 \Gamma_{2C}\,. \tag{22}$$

Additional information for these relationships is given in Section S5 in the Supporting Information.

The values for diffusivities and thermodynamic factors by MD simulations reported in data tables and figures represent the averages of three independent simulation runs. The associated uncertainties correspond to statistical uncertainties (k = 2) among the values of three independent runs.

Coordination number CN. The coordination number CN_{ij} of species j relative to species i can be calculated by integrating the radial distribution function $g_{ij}(r)$ over the radius r according to [99]

$$CN_{ij} = 4\pi\rho_j \int_0^{r'} g_{ij}(r)\, r^2 dr\,. \tag{23}$$

Here, ρ_j is the number density of species j and r' is the distance to the first minimum of $g_{ij}(r)$.

4 Results and Discussion

Within this study, ternary mixtures containing the solute [Li][NTf_2] dissolved in binary solvent mixtures of ACN and DMC or ACN and DMSO were studied by DLS experiments and MD simulations over a T range between (293.15 to 373.15) K. To study the influence of the solvent composition on the mass diffusivities, the amount fraction of the components in the binary solvent mixture is varied between 0.25, 0.50, and 0.75 while maintaining the amount fraction of [Li][NTf$_2$] x_1 in the corresponding ternary mixtures at 0.05. Furthermore, to examine the influence of the amount fraction of [Li][NTf$_2$] on the diffusive mass transport, x_1 is varied between 0.05, 0.10, 0.15, 0.20, and 0.25 in an equimolar solvent mixture of ACN and DMSO over the same T range. In the following, first, a summary of the results obtained from DLS and MD simulations is given. Afterward, the agreement between experimentally determined eigenvalues $\hat{D}_i$ and the ones obtained from MD simulations is discussed to validate the simulation results. Finally, the Fick diffusion coefficients determined by MD simulations are studied, including their kinetic and thermodynamic contributions in form of the MS diffusion coefficients and thermodynamic factors, to get a better understanding of how the intermolecular interactions and the fluid structure influence the diffusive mass transport.

4.1 Summary of Diffusivity Data

The eigenvalues $\hat{D}_i$ obtained by DLS for the ternary mixtures, where a single eigenvalue could be accessed, are presented in Table 3 together with their expanded experimental uncertainties (k = 2). $\hat{D}_i$ for mixtures, where both eigenvalues could be resolved, are listed in Table 4 together with their expanded experimental uncertainties. The expanded experimental uncertainties (k = 2) for $\hat{D}_i$ by DLS for ternary mixtures based on the solvents ACN and DMC range from (2.1 to 12)%, with an average value of 7.4%. For the ternary mixtures based on the solvents ACN and DMSO, the uncertainties vary between (1.1% and 83)% with an average of 12%. The larger uncertainties observed in the latter mixtures are attributed to the weak scattering intensity of one mass mode, which could only be resolved in mixtures with a high amount fraction of [Li][NTf_2], specifically when x_1 exceeds 0.15. For additional information, the reader is referred to Section 3.2.

The results by MD simulations for the elements of Fick diffusion coefficients matrix $\boldsymbol{D}^{\mathrm{M}}$ in the amount-averaged reference frame, as well as the eigenvalues and associated statistical uncertainties (k = 2), are presented in Tables S3 and S9 in Supporting Information. It is also important to note that the eigenvalues calculated by MD simulations yield positive and real values for all the mixtures of the present study, which is consistent with the theory of irreversible thermodynamics [52]. Furthermore, the results of the MD simulations for the elements of the matrices $\boldsymbol{\Delta}$ and $\boldsymbol{\Gamma}$ together with their statistical uncertainties (k =2) are listed in Tables S4, S5, S10, and S11 in the Supporting Information. For obtaining these matrices, DMC is considered as the reference component, i.e., component 3, for the ternary mixtures containing [Li][NTf_2], ACN, and DMC, while DMSO is selected as the reference component for ternary mixtures containing [Li][NTf_2], ACN, and DMSO. Additionally, Tables S6 and S12 in the Supporting Information contain the MS diffusion coefficients accessed by MD simulations. Self-diffusion coefficients D_i are listed in Tables S8 and S14 in the Supporting Information. The statistical uncertainties (k = 2) of the elements of the $\boldsymbol{D}^{\mathrm{M}}$ matrix for the mixtures based on ACN and DMC are on average 25%, while those for the mixtures based on ACN and DMSO are 74% on average. Here, the large relative uncertainties arise mainly from the cross-diffusion coefficient D_{12}^{M}, which exhibits much lower values compared to those of other elements. Because these values are very small, even minor changes lead to large relative uncertainties, despite the absolute uncertainty being insignificant. Nonetheless, considering its negligible magnitude, it has no impact on the data interpretation in the amount-averaged reference frame within the scope of the present work.

Table 3. Eigenvalues of the Fick diffusion Coefficient Matrix Obtained by DLS for State Points where only one Eigenvalue was Resolved Together with Their Expanded

Experimental Uncertainties (k = 2) as a Function of Temperature T at Atmospheric Pressures.[a,b]

T / K	$10^{10} \times \hat{D}_i$ / ($m^2 \cdot s^{-1}$)	$10^{10} \times U(\hat{D}_i)$ / ($m^2 \cdot s^{-1}$)
0.0499 [Li][NTf_2] (1) in 0.2500 ACN (2) and 0.7500 DMSO (3)		
293.22	9.14	0.25
322.37	15.68	0.33
347.62	22.71	0.32
372.97	31.71	0.62
0.0435 [Li][NTf_2] (1) in 0.5001 ACN (2) and 0.4999 DMSO (3)		
293.16	11.49	0.38
322.88	18.60	0.72
348.11	25.40	0.42
373.68	32.97	0.37
0.1000 [Li][NTf_2] (1) in 0.5001 ACN (2) and 0.4999 DMSO (3)		
293.08	6.98	0.34
323.02	11.99	0.51
347.43	17.11	0.58
372.69	23.62	0.74
0.1497 [Li][NTf_2] (1) in 0.5001 ACN (2) and 0.4999 DMSO (3)		
293.30	4.15	0.27
323.30	7.31	0.32
350.23	10.98	0.75
376.39	15.68	0.43
0.0370 [Li][NTf_2] (1) in 0.7500 ACN (2) and 0.2500 DMSO (3)		
293.15	11.70	0.28
322.81	17.87	0.41
347.64	24.18	0.83
372.95	31.64	0.43
0.0500 [Li][NTf_2] (1) in 0.2504 ACN (2) and 0.7496 DMC (3)		
293.11	8.23	0.97
322.87	12.05	0.72
347.38	15.8	1.49
372.61	20.39	0.75
0.0500 [Li][NTf_2] (1) in 0.4999 ACN (2) and 0.5001 DMC (3)		
293.15	13.1	1.47
323.02	17.9	1.33
347.74	22.45	0.47
372.40	28.0	1.93
0.0500 [Li][NTf_2] (1) in 0.7504 ACN (2) and 0.2496 DMC (3)		
293.13	13.9	1.20
323.16	21.43	0.74

347.77	27.5	1.82
373.00	34.7	4.23

[a]The amount fractions of the components were determined gravimetrically during mixture preparation using a balance with an expanded uncertainty of 1 mg at a 95% confidence level (k = 2). [b]Expanded uncertainty for T $U(T)$ = 0.050 K (k = 2). The reported values of T represent mean values, averaged over the entire measurement period at each investigated state point.

Table 4. Eigenvalues of Fick diffusion Coefficient Matrix Obtained by DLS for State Points where Both Eigenvalues were Resolved Together with Their Expanded Experimental Uncertainties (k = 2) for Different Solvent and Solute Amount Fractions[a] as a Function of Temperature T at Atmospheric Pressure.[a,b]

T / K	$10^{10} \times \hat{D}_1$ /(m^2·s^{-1})	$10^{10} \times U(\hat{D}_1)$ / (m^2·s^{-1})	$10^{10} \times \hat{D}_2$ / (m^2·s^{-1})	$10^{10} \times U(\hat{D}_2)$ / (m^2·s^{-1})
0.2000 [Li][NTf$_2$] (1) in 0.5009 ACN (2) and 0.4991 DMSO (3)				
293.14	2.39	0.21		
322.81	4.66	0.35	1.02	0.37
347.37	7.31	0.56	2.2	1.83
372.39	9.98	0.80	4.14	0.77
0.2500 [Li][NTf$_2$] (1) in 0.4996 ACN (2) and 0.5004 DMSO (3)				
293.39	1.45	0.64	0.28	0.05
323.28	3.07	0.44	0.65	0.14
348.02	4.7	1.20	1.18	0.40
373.13	8.5	1.58	2.64	0.52

[a]The amount fractions of the components were determined gravimetrically during mixture preparation using a balance with an expanded uncertainty of 1 mg at a 95% confidence level (k = 2). [b]Expanded uncertainty for T $U(T)$ = 0.050 K (k = 2). The reported values of T represent mean values, averaged over the entire measurement period at each investigated state point.

4.2 Results for the Eigenvalues of the Fick Diffusion Coefficient Matrix by DLS and MD Simulations

Within this section, the eigenvalues $\hat{D}_i$ of the Fick diffusion coefficient matrix obtained from DLS and MD simulations are discussed. Here, it should be mentioned again that the eigenvalues are independent of the reference frame so that a direct comparison between the results from DLS and MD simulations is possible. In the case that only one eigenvalue could be accessed by DLS, it should also be determined whether this value represents the larger or the smaller eigenvalue. For this, the results by DLS for the binary subsystems are considered where appropriate. Finally, the agreement between the eigenvalues from DLS and MD simulations is presented.

Ternary mixtures of [Li][NTf$_2$], ACN, and DMC. In Figure 5, $\hat{D}_i$ obtained by DLS for mixtures containing 0.05 [Li][NTf$_2$](1) in binary solvent mixtures of x_2'·ACN (2) and (1–x_2')·DMC (3) at x_2' = (0.25, 0.50, and 0.75) are presented as a function of T. In addition, D_{11} for the binary subsystems 0.05 [Li][NTf$_2$] + 0.95 ACN, 0.05 [Li][NTf$_2$] + 0.95 DMC, and 0.25 ACN + 0.75 DMC are included in the figure.

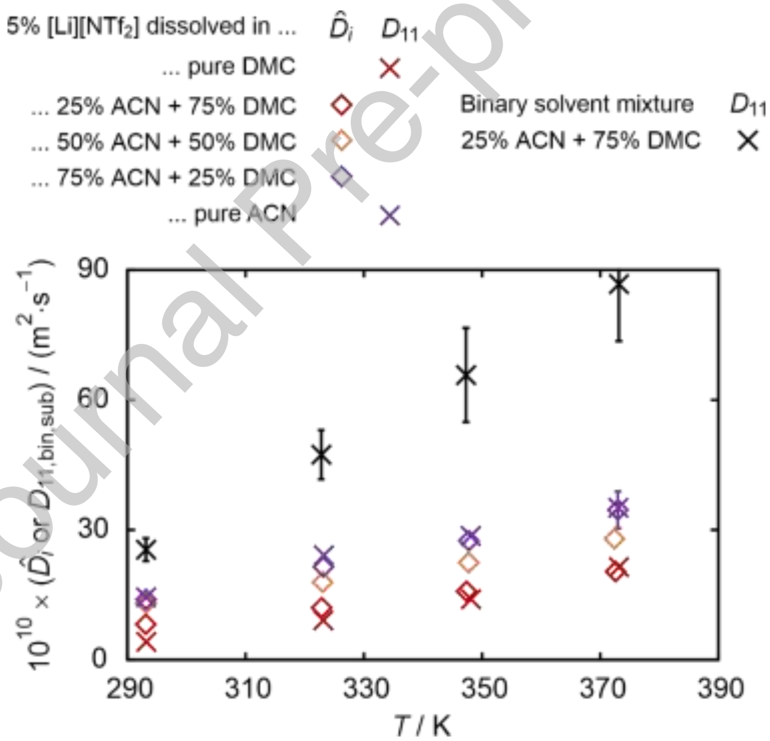

Figure 5. The eigenvalue of the Fick diffusion coefficient matrix accessed by DLS (diamonds) for the ternary mixtures containing 0.05 [Li][NTf$_2$] dissolved in binary solvent mixtures of x_2'·ACN and (1–x_2')·DMC at x_2' = (0.25, 0.50, or 0.75) and as a function of T. The Fick diffusion coefficients D_{11} (crosses) by DLS for binary subsystems consisting of 0.05 [Li][NTf$_2$] + 0.95 ACN (purple), 0.05 [Li][NTf$_2$] + 0.95 DMC (red), and 0.25 ACN + 0.75 DMC

DMSO (black) are included for comparison. Data for the binary subsystems consisting of $[Li][NTf_2]$ are taken from previous publications [37,38].

$\hat{D}_i$ increases with increasing T for all three ternary mixtures, likely due to the enhancement of system dynamics with increasing T. In addition, the eigenvalues for all three mixtures lie between the Fick diffusion coefficients D_{11} of the binary subsystems containing $0.05[Li][NTf_2]$ + 0.95ACN, and $0.05[Li][NTf_2]$ + 0.95DMC. The eigenvalues for the ternary mixture based on the solvent mixture containing 75% DMC are mostly within combined uncertainties with the binary subsystems based on DMC. With increasing amount fraction of ACN, $\hat{D}_i$ converge to D_{11} of binary subsystem of $0.05[Li][NTf_2]$ + 0.95 ACN. These results indicate qualitatively that the eigenvalue obtained from DLS for this system is primarily influenced by the diffusion of $[Li][NTf_2]$ and can be approximated by a pseudo binary mixture consisting of the lithium salt and the pseudo pure component representing the solvent mixture. From the above observations, it is, however, not clear whether the eigenvalue accessed by DLS reflects the lower or the higher one. To clarify this, the results for the eigenvalues by DLS are compared to those obtained by MD simulations in the following. In Figure 6, the corresponding results by DLS and MD simulations are presented for the mixture containing 0.05 $[Li][NTf_2]$ dissolved in a binary solvent mixture of 0.25 ACN + 0.75 DMC as a function of T.

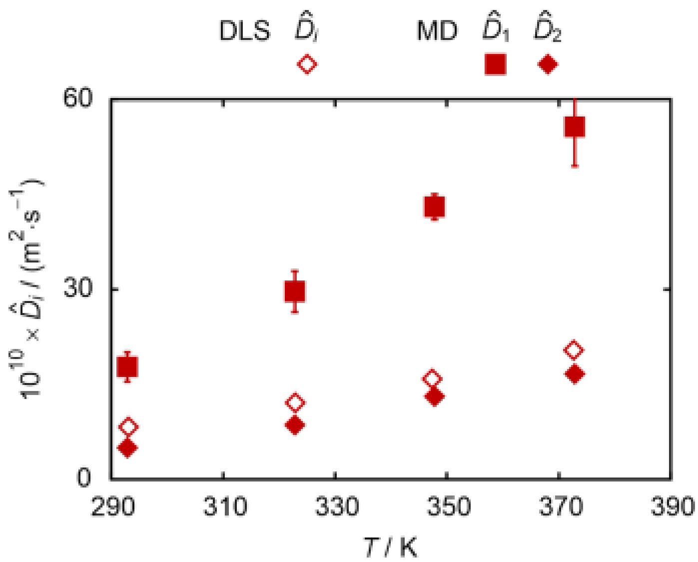

Figure 6. Eigenvalues accessed by DLS (open diamonds) and MD simulations (filled diamonds and squares) for the ternary electrolyte mixture containing 0.05 [Li][NTf_2] dissolved in a binary solvent mixture of 0.25 ACN + 0.75 DMC as a function of T.

Here, the lower eigenvalue $\hat{D}_2$ obtained from MD simulations agrees well with the eigenvalue accessed by DLS, indicating that DLS accesses the lower eigenvalue. This observation is consistent with Bardow's analysis [27] of DLS results in ternary mixtures based on organic liquids with low molecular weights, such as the mixture of glycerol, acetone, and water.

Ternary mixtures of [Li][NTf_2], ACN, and DMSO. Figure 7 depicts $\hat{D}_i$ by DLS for the ternary mixtures containing 0.05[Li][NTf_2] dissolved in x_2'·ACN (2) and (1-x_2')·DMSO (3) at x_2' = (0.25, 0.50, and 0.75) as a function of T. For comparison, D_{11} by DLS for the binary subsystems consisting of 0.05 [Li][NTf_2] + 0.95 ACN, 0.05 [Li][NTf_2] + 0.95 DMSO, and 0.25 ACN + 0.75 DMSO are also included.

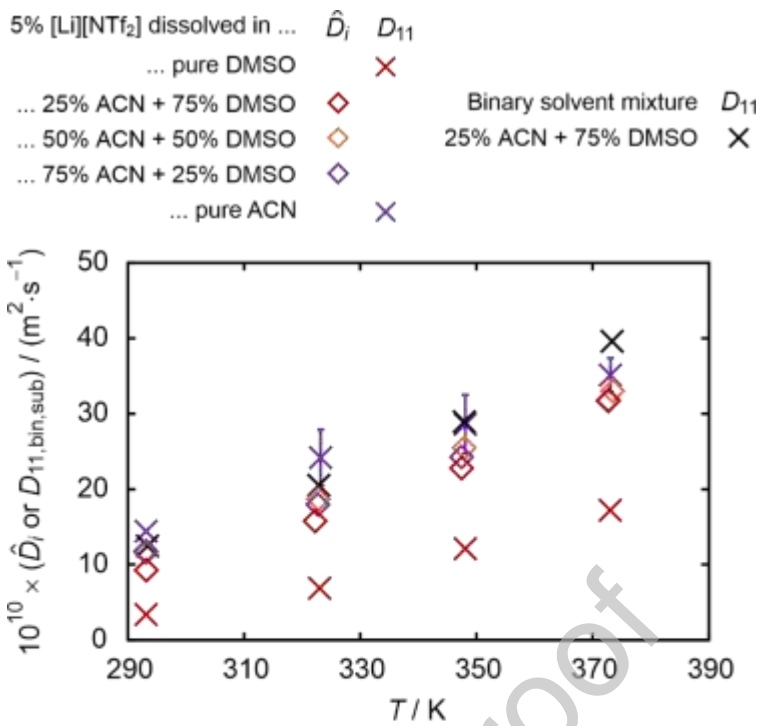

Figure 7. Eigenvalues (diamonds) accessed by DLS for ternary mixtures containing 0.05 $[Li][NTf_2]$ dissolved in binary solvent mixtures of x_2'·ACN and $(1–x_2')$·DMSO at x_2' = (0.25, 0.50, and 0.75) as a function of T. The Fick diffusion coefficients D_{11} (crosses) by DLS for binary subsystems consisting of 0.05 $[Li][NTf_2]$ + 0.95 ACN, 0.05 $[Li][NTf_2]$ + 0.95 DMSO, and 0.25 ACN + 0.75 DMSO are also represented. Data for the binary subsystems consisting of $[Li][NTf_2]$ are taken from previous publications [37,38].

As discussed for the previous ternary mixtures, $\hat{D}_i$ increases with increasing T. Unlike for ternary mixtures composed of DMC, however, no clear dependence of resolved $\hat{D}_i$ on the solvent composition can be identified. The eigenvalues of all three ternary mixtures are in the vicinity of D_{11} of binary subsystems based on ACN. Unlike for the ternary mixtures containing DMC, D_{11} of the binary solvent mixture also falls within the range of the obtained $\hat{D}_i$. Because of this, it is not possible to interpret whether the obtained eigenvalue might be estimated with the help of a pseudo binary subsystem, as it was done for the mixtures containing DMC. To clarify whether DLS yields the higher or lower eigenvalue for the mixtures with low salt amounts, the results for mixtures containing different $x_{[Li][NTf2]}$ in an equimolar mixture of ACN and DMSO are shown in Figure 8.

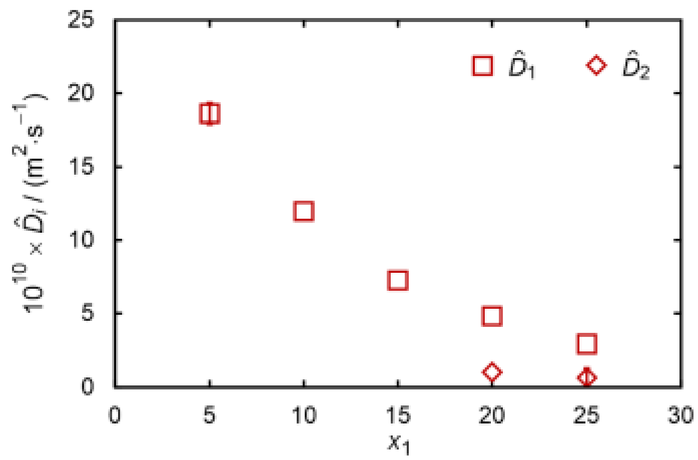

Figure 8. Eigenvalues by DLS for the ternary mixtures containing [Li][NTf$_2$] dissolved in equimolar binary mixture of ACN and DMSO as a function of amount fraction of the salt x_1 at T = 323.15 K. The higher and lower eigenvalues are shown by square and diamond symbols, respectively.

The results show that, at high $x_{[Li][NTf2]}$, both eigenvalues are accessible by DLS. This also shows that the eigenvalue accessible at $x_{[Li][NTf2]}$ = 0.05 is most probably the larger of the two eigenvalues. The general trend of the decreasing eigenvalues with increasing x_1 can be attributed to increasing viscosities of the mixtures with the addition of the salt, as well as to changes in the ion dissociation, which will be discussed with the help of MD simulation in section 4.4. In Figure 9, the results for the larger eigenvalue accessible by DLS for ternary mixtures based on an equimolar solvent mixture are shown as a function of $x_{[Li][NTf2]}$ for different temperatures. Here, the results show that the eigenvalues converge to D_{11} of the binary subsystem consisting of an equimolar mixture of ACN and DMSO as $x_{[Li][NTf2]}$ converges to zero.

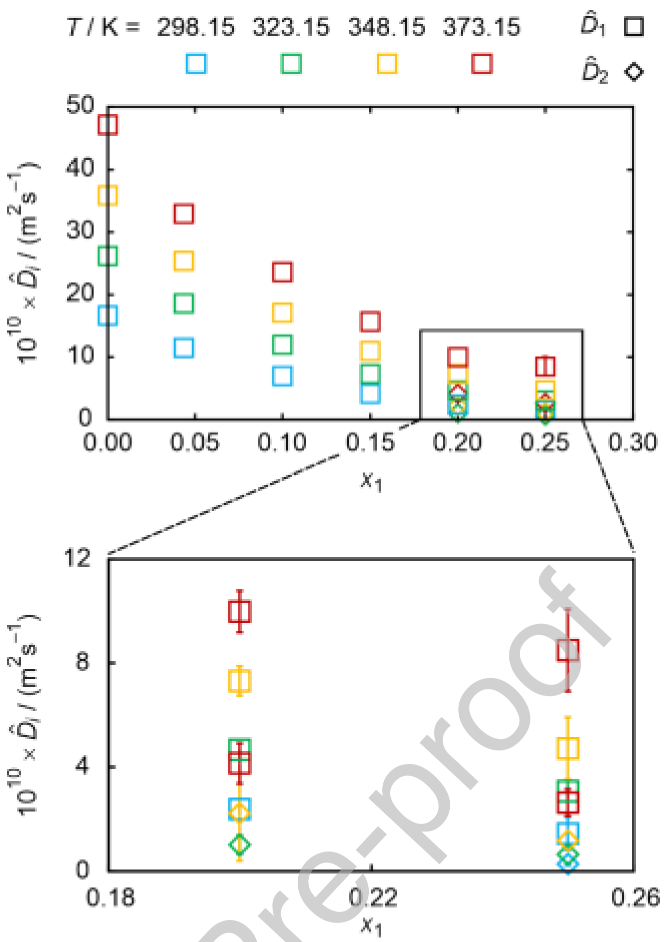

Figure 9. Eigenvalues by DLS for ternary mixtures consisting of [Li][NTf_2] dissolved in an equimolar mixture of ACN and DMSO as a function of amount fraction of the salt x_1 at different temperatures. The higher and lower eigenvalues are shown by square and diamond symbols, respectively. Error bars smaller than the symbol sizes are not shown.

Comparison between the eigenvalues obtained by DLS and MD simulations. Since DLS provides access to the eigenvalues of the Fick diffusion coefficient matrix, a comparison between the results by DLS and MD simulations is only possible through a comparison of the eigenvalues determined by both methods. Such a comparison is valid since the eigenvalues of the Fick diffusion coefficient matrix are independent of the reference frame. As discussed previously, in cases where only a single eigenvalue was resolvable by DLS, the technique accesses the lower eigenvalue for mixtures consisting of [Li][NTf_2], ACN, and DMC and the higher eigenvalue for those containing [Li][NTf_2], ACN, and DMSO. The reason why DLS resolves only a single eigenvalue for certain mixtures is explained in Section 3.2.

Consequently, for corresponding mixtures, the eigenvalues obtained from MD simulations are assigned in the same order to facilitate the comparison. The agreement between the results from DLS and MD simulations is illustrated in Figure 10 in form of a parity plot. The results by MD simulations agree with the results from DLS with an AARD of 16% for the mixtures based on ACN and DMC and 24% for those based on binary mixtures of ACN and DMSO. The results show that MD simulations are able to predict the diffusivities in ternary electrolyte systems. Here, it should be mentioned again that simulated results for the ternary mixtures are purely predictive and that no interaction parameters were used.

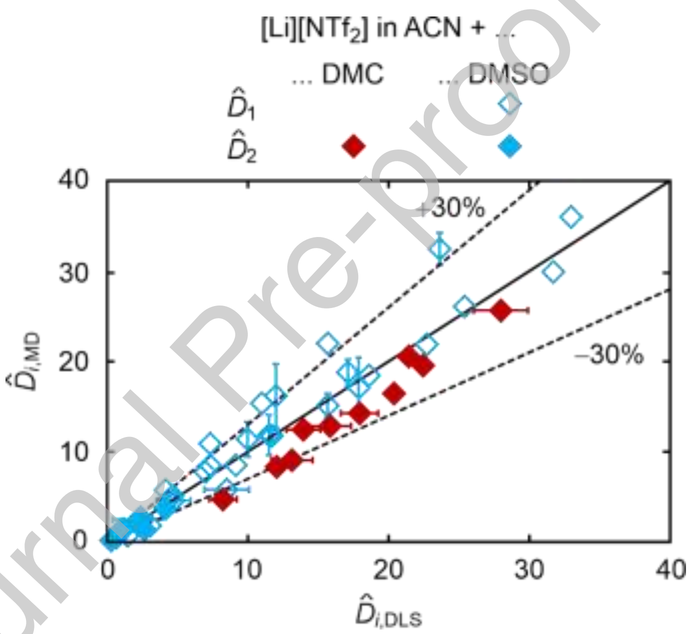

Figure 10. Comparison between the eigenvalues $\hat{D}_i$ by DLS and MD simulations for ternary mixtures containing [Li][NTf$_2$] dissolved in binary mixtures of ACN and DMC (red symbols) or ACN and DMSO (blue symbols). For a given state point, $\hat{D}_1$ and $\hat{D}_2$ represent the higher eigenvalue (open symbols) and lower eigenvalue (filled symbols), respectively. The line represents a perfect agreement between the two data sets. Error bars smaller than symbol sizes are not shown.

4.3 Influence of the Solvent Composition on Fick Diffusion Coefficients by MD Simulations

Within this section, the influence of the solvent composition on the elements of $\boldsymbol{D}^{\mathrm{M}}$ should be discussed. For a discussion about the influence of temperature on the different elements of $\boldsymbol{D}^{\mathrm{M}}$, the reader is referred to section S10 in the Supporting Information. As shown there, the diagonal elements, which have positive values, increase with increasing T, as can be seen for the Fick diffusion coefficients of the binary mixtures. The off-diagonal elements, on the other hand, can be either positive or negative. In most cases, their magnitude, regardless of the sign, also increases with increasing T.

Ternary electrolyte mixtures containing [Li][NTf$_2$] (1), ACN (2), and DMC (3). In the upper part of Figure 11, the elements of the Fick diffusion coefficient matrix are shown for mixtures containing 0.05[Li][NTf$_2$] (1) dissolved in a binary mixture of x_2'·ACN (2) + $(1-x_2')$·DMC (3) as a function of x_2' at T = 293.15 K. The diagonal elements show positive values, which agree with observations reported in the literature for multicomponent mixtures [43,49]. Among the elements of the Fick diffusion coefficient matrix, the largest values are given for D_{22}^{M}, followed by D_{11}^{M}. The diagonal element D_{11}^{M}, which describes the diffusive amount flux of [Li][NTf$_2$] (1) with respect to its own amount fraction gradient, increases with increasing x_2'. This agrees with the observations for the binary subsystems of [Li][NTf$_2$] dissolved in either ACN or DMC [38], where a larger D_{11} was observed for the mixture based on ACN. D_{22}^{M}, which links the diffusive amount flux of ACN (2) due to its own amount fraction gradient, shows a slight enhancement from x_2' = (0.25 and 0.50) and remains unchanged afterward. This trend closely resembles the composition-dependent trend of the Fick diffusion coefficient in the binary subsystem of ACN (2) + DMC (3), c.f. Supporting Information.

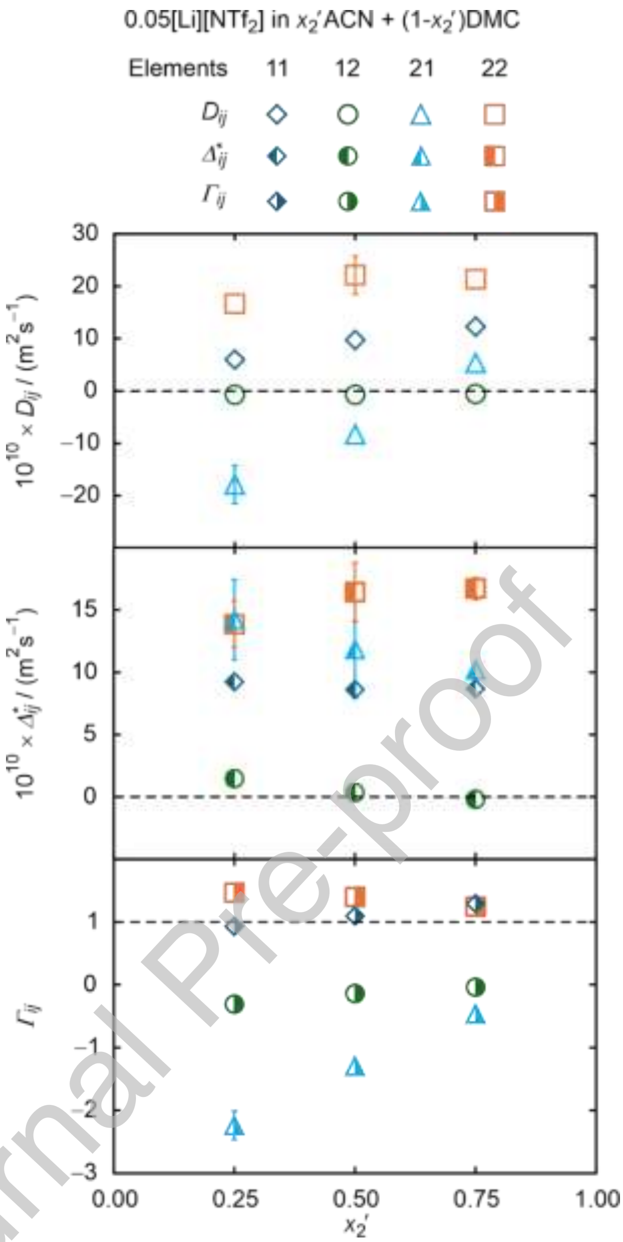

Figure 11. Elements of matrices $\boldsymbol{D}^{\mathrm{M}}$ (top), $\boldsymbol{\Delta}^{*}$ (middle) and $\boldsymbol{\Gamma}$ (bottom) by MD simulations for the ternary mixtures containing 0.05 [Li][NTf$_2$] dissolved in binary mixtures of solvents x_2'ACN + (1 − x_2')DMC as a function of x_2' at T = 293.15 K. Error bars smaller than symbol sizes are not shown.

The off-diagonal element D_{12}^{M}, describing the diffusive amount flux of [Li][NTF$_2$](1) due to an amount fraction gradient of ACN(2), remains nearly unchanged over the entire composition range and is approximately one order of magnitude smaller than the diagonal elements. This suggests that the diffusive amount flux of the $[Li][NTf_2](1)$ is predominantly governed by its own amount fraction gradients. D_{21}^{M} initially shows negative values, where its magnitude decreases up to $x_2' = 0.50$. With further increasing x_2', it shows a sign crossover from negative to positive values. Additionally, the magnitude of D_{21}^{M} is in the same order as the diagonal elements, indicating strong coupling effects. The observed trends for D_{21}^{M} suggest that ACN tends to diffuse toward $[Li][NTf_2]$-rich regions in mixtures with low amount of ACN. This behavior can be explained by the role of ACN in the dissociation of lithium salt. As was shown in our previous work, ACN plays an important role in the dissociation of the lithium cation and on average 4 ACN molecules can be found in the solvation shell surrounding the cation [37,38]. Due to much smaller D_{12}^{M} values, the resulting Fick diffusion coefficient matrix tends to resemble a lower triangular matrix, i.e., the D_{12} entry is close to zero. For such a matrix, the eigenvalues should be identical to the diagonal elements of the corresponding matrix. This is given for the mixture at hand, where the eigenvalues are within the combined uncertainties of the diagonal elements, c.f. Section S9 in the Supporting Information.

To gain further insights into how the kinetic contribution and the fluid structure influence the Fickian diffusion process, the latter can be decoupled into the kinetic and thermodynamic contributions. The kinetic contribution is contained in $\boldsymbol{\Delta}^*$, which contain the individual MS diffusion coefficients, while the thermodynamic contribution is given by the matrix of thermodynamic factors $\boldsymbol{\Gamma}$. The elements of $\boldsymbol{\Delta}^*$ and $\boldsymbol{\Gamma}$ are displayed in the middle and lower panel of Figure 11 as function of x_2' and at $T = 293.15$ K. The magnitude of Δ_{12}^{*} is close to zero across the entire solvent composition range. This highlights its negligible contribution to the Fick diffusion coefficients. All the other Δ_{ij}^{*} values are positive. Among them, Δ_{22}^{*} tends to exhibit the largest values across all compositions, followed by Δ_{21}^{*} and Δ_{22}^{*}. While the diagonal elements Δ_{ii}^{*} show a negligible dependency on x_2', Δ_{21}^{*} first overlaps with Δ_{22}^{*} at x_2' = 0.25, after which its magnitude reduces with increasing x_2' and tends to converge to Δ_{11}^{*} at x_2' = 0.75. The analysis of Δ_{ij}^{*} shows that negative Fick diffusion coefficients do not originate from the kinetic contribution. Therefore, the negative values of D_{ij} arise from the thermodynamic contribution, which is discussed in the following.

In general, the departure of Γ_{ii} from unity and off-diagonal elements $\Gamma_{ij,i\neq j}$, which are not close to zero, indicate a strongly non-ideal fluid structure of these mixtures. Γ_{22} remains larger than unity over the investigated composition range and tends to reduce with increasing x_2'. This indicates that the distinct interactions between ACN and the solute ions facilitate the diffusive mass transport, most probably due to the dissociation of the ions. This will also be discussed in more detail in the following with the help of *CN*s obtained by MD simulations. On the other hand, both Γ_{11} and Γ_{12} imply their ideal contributions, with Γ_{11} values close to unity and Γ_{12} values close to zero in most cases. However, Γ_{21} shows significant negative values over the entire investigated composition range. This indicates that the chemical potential of ACN decreases within the vicinity of [Li][NTf_2] and implies that ACN is attracted towards [Li][NTf_2] rich regions, especially in mixtures with low ACN amount fractions, i.e., low x_2' values. However, this effect appears to reduce with increasing x_2'. These observations for the thermodynamic factors are consistent with the trends discussed for the Fick diffusion coefficients, particularly for D_{21}^{M}, and imply that the composition-dependent trend of the Fick diffusion coefficients is mainly governed by the thermodynamic contribution. To gain further insights, the fluid structure, specifically the dissociation of the ions, is investigated in the following.

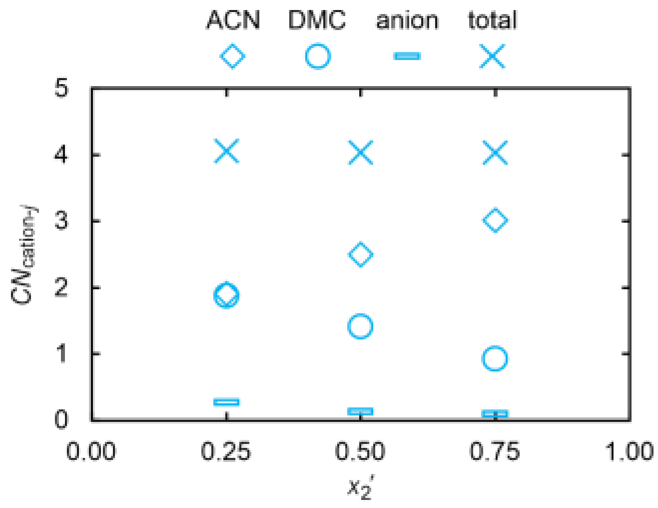

Figure 12. Coordination numbers CN_{ij} of ACN ($CN_{\text{cation-ACN}}$), DMC ($CN_{\text{cation-DMC}}$) and anion ($CN_{\text{cation-anion}}$) around a cation by MD simulations for the ternary mixtures containing 0.05 [Li][NTf_2] dissolved in binary mixtures of solvents x_2'ACN + (1 – x_2')DMC as a function of x_2' at T = 293.15 K. Error bars are not shown as they are smaller than the symbol sizes.

To investigate the arrangement of molecules surrounding the ions, the CNs between $[Li]^+$ and ACN, DMC or anion were calculated. These CNs were obtained by integrating the corresponding RDFs, as outlined in section 3.3. The determined CNs are depicted in Figure 12 for these mixtures as a function x_2' at T = 293.15 K. Despite the composition of the solvents, Figure 12 illustrates that $[Li]^+$ solvation shell retains a four-site coordination structure. Furthermore, results suggest that the $CN_{\text{cation-anion}}$ shows a decreasing trend with increasing x_2'. Nonetheless, these values are much smaller than the other CNs, indicating a higher ion dissociation probability in the mixtures. Similar observations for $CN_{\text{cation-anion}}$ with increasing the amount of solvent with the higher ε_r over the solvent with lower ε_r have been reported for ternary electrolyte mixtures containing low content of [Li][PF_6] dissolved in EC and DMC solvents by Kiyobayashi et el. [9] using MD simulations. For latter mixtures with a low salt concentration, Borodin et al. [36] also reported a lower probability of finding an anion in the first solvation shell of a cation using Born Oppenheimer MD simulations. At x_2'

= 0.25, the solvation shell surrounding $[Li]^+$ is found to contain nearly equal amounts of ACN and DMC molecules, despite the low amount of ACN in the mixture. This suggests preferential coordination of $[Li]^+$ by ACN over the DMC, which could be due to high polarity of ACN compared to DMC as it is also reflected in their ε_r values, c.f. Table 1. This preferential arrangement of ACN surrounding the lithium cation is likely the cause for the negative Γ_{12}, as discussed above. With increasing x_2', $CN_{\text{cation-ACN}}$ increases while $CN_{\text{cation-DMC}}$ decreases. Previous studies on ternary electrolyte mixtures have also reported that increasing the bulk composition of the solvent with a larger ε_r leads to an increase in its composition or CN in the first solvation shell of the cation [9]. Nonetheless, a direct comparison with them is not possible due to the difference in the mixtures investigated. At $x_2' = 0.75$, approximately 3 ACN molecules and 1 DMC molecule can be found within the solvation shell surrounding $[Li]^+$. Since, however, the composition of the solvents in the solvation shell is the same as the overall composition of the solvent molecules in the mixture, this does not represent an enrichment of ACN in comparison to DMC, which can explain the reduced magnitude of the negative Γ_{21} values with increasing x_2'.

Ternary electrolyte mixtures containing [Li][NTf$_2$] (1), ACN (2), and DMSO (3). In Figure 13, the elements of $\boldsymbol{D}^{\mathrm{M}}$, $\boldsymbol{\Delta}^*$, and $\boldsymbol{\Gamma}$ for the mixture containing 0.05 [Li][NTf$_2$] (1) dissolved in x_2'·ACN (2) + $(1-x_2')$·DMSO (3) at x_2' = (0.25, 0.50 and 0.75) and at T = 293.15 K are shown. Like for the mixture based on ACN and DMC, both diagonal elements of $\boldsymbol{D}^{\mathrm{M}}$ show positive values. Here, D_{11}^{M} increases with increasing x_2', which can be explained by a larger D_{11} of [Li][NTf$_2$] in ACN in comparison to DMSO [38]. D_{21}^{M} and D_{22}^{M} show an increase from $x_2' = 0.25$ to $x_2' = 0.50$. Afterward, both coefficients remain unchanged with further increasing x_2'. D_{12}^{M} remains approximately an order of magnitude smaller than the diagonal elements across all compositions, which confirms the previously mentioned statement that the diffusive mass transport of [Li][NTf$_2$] appears to be predominantly governed by its own concentration gradients. Unlike the previously discussed mixture containing DMC, all elements of the Fick diffusion coefficient matrix are positive, within the statistical uncertainties, for this mixture.

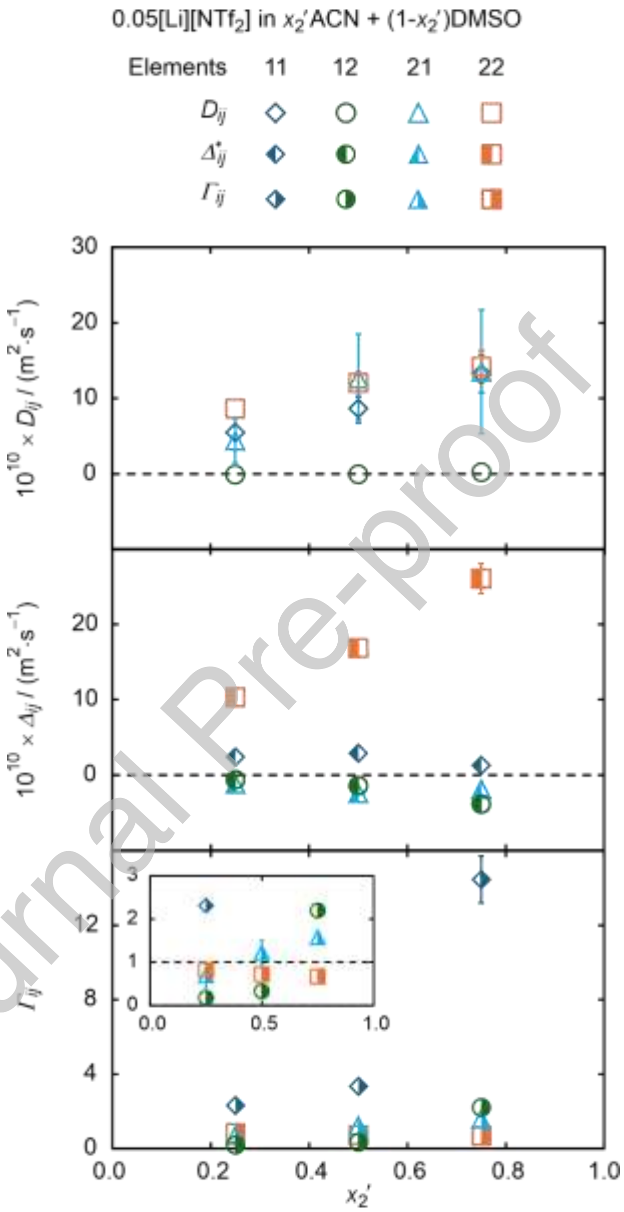

Figure 13. Elements of matrices $\boldsymbol{D}^{\mathrm{M}}$ (top), $\boldsymbol{\Delta}^{*}$ (middle) and $\boldsymbol{\Gamma}$ (bottom) by MD simulations for the ternary mixtures containing 0.05 [Li][NTf_2] dissolved in binary mixtures of solvents x_2'ACN + (1 – x_2') DMSO as a function of x_2' at T = 293.15 K. Error bars smaller than symbol sizes are not shown.

With respect to the elements of Δ^*, Δ^*_{22} exhibits the largest values across the entire composition range. Furthermore, Δ^*_{22} increases with increasing x_2' and shows a strong composition-dependency. This trend appears to arise not only from the apparent decrease in viscosity with increasing x_2', but also from the strong non-ideal interactions between ACN and DMSO, as discussed in the Supporting Information. Δ^*_{11}, on the other hand, increases slightly up to x_2' = 0.50 and then tends to decrease with further increasing x_2'. In contrast to the diagonal elements, both Δ^*_{12} and Δ^*_{21} are negative for all compositions and increase in their magnitude with increasing x_2', meaning that both [Li][NTf_2] and ACN diffuse against the chemical potential gradients of each other.

The mixtures show significant deviations of Γ_{ii} from unity as well as non-negligible $\Gamma_{ij,i\neq j}$ values, highlighting the presence of strong non-ideal interactions. Γ_{11} exhibits the largest value which is clearly above unity. It increases slightly up to x_2' = 0.50 and then exhibits a pronounced rise with further increasing x_2', reaching a value of 14.69 at x_2' = 0.75. Such high values of the thermodynamic factors have been reported in previous studies on binary electrolyte mixtures containing solvents with high dielectric constants, albeit at higher salt concentrations [100]. For legibility, the inset in the bottom panel of Figure 13 depicts the dependency of Γ_{12}, $\Gamma_{21,}$ and Γ_{22} on x_2'. Γ_{22} values are positive but below unity. It shows a decrease with increasing x_2' from a value of 0.82 at x_2' = 0.25 to a value of 0.67 at x_2' = 0.75. The reduction of Γ_{22} with increasing x_2' might be attributed to a microscopic structure formation, such as the formation of ACN clusters in the solution due to unfavorable interactions between ACN and DMSO. This was also observed in the thermodynamic factors of the binary solvent subsystem, c.f. Section S11 in Supporting Information. Both Γ_{12} and Γ_{21} increase with increasing x_2'. In general, the non-ideal effects of these mixtures appear to be rather complicated and more pronounced than those observed in the previously discussed

DMC-based systems, most likely due to strong non-ideal interactions between ACN and DMSO. To further investigate these interactions, the solvation structure is discussed in the following with the help of *CN*s.

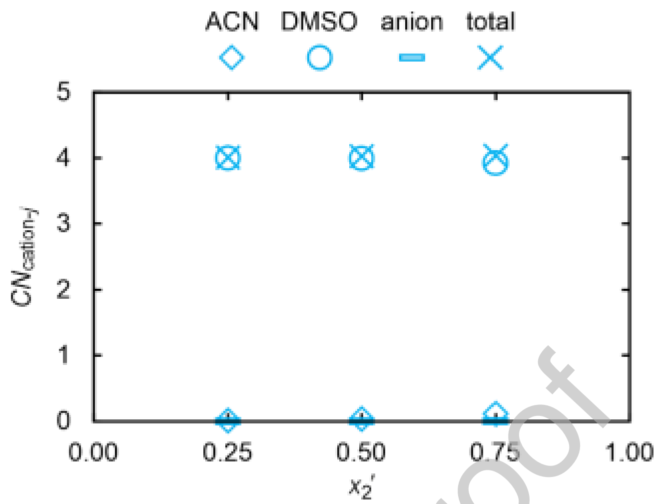

Figure 14. Coordination numbers CN_{ij} of ACN ($CN_{\text{cation-ACN}}$), DMC ($CN_{\text{cation-DMSO}}$) and anion ($CN_{\text{cation-anion}}$) around a cation by MD simulations for the ternary mixtures containing 0.05 [Li][NTf_2] dissolved in binary mixtures of solvents x_2'ACN + (1 – x_2')DMSO as a function of x_2' at T = 293.15 K. Error bars are not shown as they are smaller than the symbol sizes.

Similar to what was done for the ternary mixtures containing DMC, the composition of molecules surrounding Li^+ cations was resolved through the calculation of *CN*s via radial distribution functions (RDFs). These results are shown in Figure 14 for the ternary mixtures under consideration as a function of x_2' at T = 293.15 K. $CN_{\text{cation-anion}}$ shows very negligible values close to zero compared to other *CN*s. This suggests an even higher degree of salt dissociation in these mixtures compared to those with DMC and ACN as solvents. Furthermore, unlike DMC-based mixtures, where the solvation shell was occupied by ACN and DMC molecules, the present system shows that the solvation shell is occupied exclusively by DMSO molecules. This agrees with the observations reported in previous studies, which investigated mixtures consisting of ACN, DMSO, and lithium hexafluorophosphate [101]. This behavior can be attributed to the higher polarity of DMSO compared to ACN, as well as unfavorable interactions between solvent molecules. The resulting solvation structures are quite stable, and cations and anions are found to be completely dissociated by the DMSO molecules.

4.4 *Influence of the Solute Composition on the Fick Diffusion Coefficient*

To study the influence of the solute composition on the Fick diffusion coefficients, ternary mixtures containing [Li][NTf_2] dissolved in an equimolar mixture of ACN and DMSO were investigated at different amount fraction of [Li][NTf_2] ranging from 0.05 to 0.25. In the top panel of Figure 15, the results for the elements of $\boldsymbol{D}^{\mathrm{M}}$ are shown as a function of x_1 at T = 293.15 K. Both diagonal elements, D_{11}^{M} and D_{22}^{M}, show positive values and decrease with increasing x_1, likely due to the increasing viscosity of the mixtures. While D_{22}^{M} initially exhibits larger values than D_{11}^{M} at the lowest investigated x_1, the two coefficients cross at x_1 = 0.10, beyond which D_{11}^{M} becomes larger. The increase in D_{11}^{M} relative to D_{22}^{M} seems to be due to a complex interplay between kinetic and thermodynamic effects, which are discussed in more detail later in this section. While D_{22}^{M} decreases asymptotically with increasing x_1, D_{11}^{M} shows a more pronounced slowing down at x_1 > 0.15, likely due to the formation of ion pairs or clusters, indicating the approaching solubility limit.

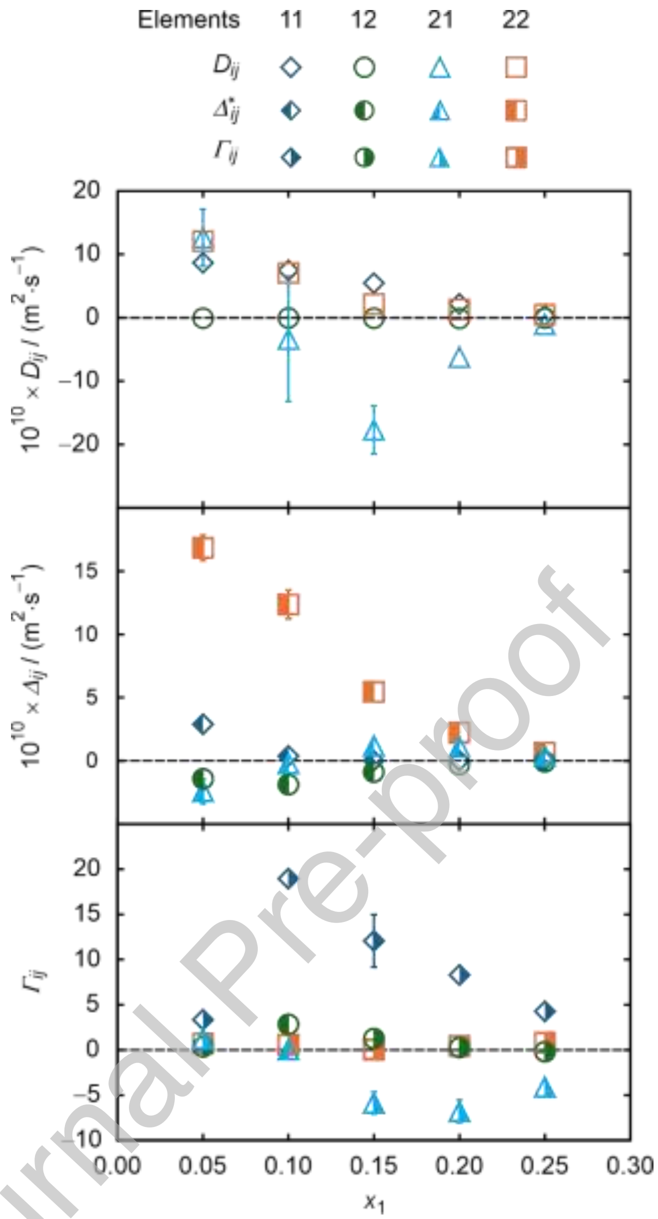

Figure 15. Elements of the matrices $\boldsymbol{D}^{\mathrm{M}}$ (top), $\boldsymbol{\Delta}^*$ (middle), and $\boldsymbol{\Gamma}$ (bottom) by MD simulations for the ternary mixtures containing [Li][NTf_2] dissolved in equimolar binary mixtures of ACN and DMSO as a function of amount fraction of the salt x_1 at T = 293.15 K. Error bars smaller than symbol sizes are not shown.

With respect to the off-diagonal elements of $\boldsymbol{D}^{\mathrm{M}}$, D_{12}^{M} shows no significant composition dependency and is approximately an order of magnitude smaller than the diagonal elements. This again implies that the diffusive amount flux of [Li][NTf_2] is primarily driven by its own amount fraction gradients. These findings may further support the applications, which focus on mass transport of lithium salt [6]. D^{M}_{21} shows a strong non-ideal behavior. It is initially positive up to x_1 = 0.10, after which it becomes negative. This indicates that ACN becomes more involved in the dissociation of the lithium cation at concentrations above x_1 = 0.10 and diffuses towards the cation. This observation will be further collaborated with the help of the obtained *CN*s in the following. Above x_2 = 0.15, the magnitude of the negative D^{M}_{21} value decreases asymptotically with further increasing x_2. These observed trends for the Fick diffusion coefficients appear to result from the interplay between kinetic and thermodynamic effects, which are contained in the matrices Δ^* and Γ, and which will be discussed in the following.

The elements of Δ^* are illustrated in the middle panel of Figure 15 as a function of x_1 at T = 293.15 K. Among all four elements, the diagonal element Δ^*_{22} exhibits the largest value across the entire composition range. Furthermore, both diagonal elements, Δ^*_{11} and Δ^*_{22}, are positive. While Δ^*_{11} decreases steadily with increasing x_1, Δ^*_{22} shows a sharp reduction up to x_1 = 0.10, beyond which no clear composition-dependent trend could be observed outside uncertainties. The off-diagonal element Δ^*_{12} remains negative throughout the entire composition range, and its magnitude generally decreases with increasing x_1. On the other hand, Δ^*_{21} initially exhibits negative values, with its magnitude decreasing up to x_1 = 0.10. Beyond this point, it undergoes a sign change to positive values. After x_1 = 0.10, it shows a decrease with further increasing x_1. In general, all four Δ^*_{ij} tend to decrease with increasing x_1, which can be attributed to increasing mixture η with increasing x_1.

The elements of the matrix Γ are shown in the bottom part of Figure 15. The values further highlight the strong non-ideal behavior of the mixtures. Γ_{11} remains positive throughout the entire composition range and increases up to a value of about 19 at x_1 = 0.10. Beyond x_1 = 0.10, Γ_{11} decreases sharply with increasing x_1, likely due to the formation of ion pairs at higher solute concentrations. Γ_{22} is positive and decreases from a value of 0.72 at x_1 = 0.05 to a minimum value of 0.08 at x_1 = 0.15. Afterward, it increases with further increasing x_1. Γ_{12} shows a similar trend to Γ_{11} but with lower magnitudes, reaching 2.8 at x_1 = 0.10 before decreasing. On the other hand, Γ_{21} is initially positive but becomes negative with increasing x_1. The magnitude of its negative values increases up to x_1 = 0.20, after which it shows a decrease. These negative values of Γ_{21} at these compositions suggest preferential interactions between ACN and the salt components. The results show that the influence of the solute composition on the kinetic and thermodynamic contributions of the Fick diffusion coefficients—and their resulting interplay—appears to be more intricate than in the case of solvent composition dependence.

The result of the product $\Delta^{*}_{11}\Gamma_{11} + \Delta^{*}_{12}\Gamma_{21}$ contributes to D^{M}_{11}. Up to x_1=0.10, $\Delta^{*}_{11}\Gamma_{11}$ is positive and becomes the dominant term in D^{M}_{11}. Beyond this point, its contribution diminishes mostly due to smaller values of Δ^{*}_{11}, regardless of Γ_{11}. Conversely, the product of $\Delta^{*}_{12}\Gamma_{21}$ becomes positive, although both individual elements Δ^{*}_{12} and Γ_{21} are negative, and becomes the main contribution to D^{M}_{11}, primarily due to the magnitude of Γ_{21}. This observation is interesting, since the cross coefficients Δ^{*}_{12} and Γ_{21} are the main contributors for D^{M}_{11}, even though the diffusive mass transport of [Li][NTf_2] is governed predominantly by its own concentration gradient. Overall, the collective contribution exhibits a smooth and decreasing trend for D^{M}_{11} with increasing x_1. For D^{M}_{12}, $\Delta^{*}_{11}\Gamma_{12}$ and $\Delta^{*}_{12}\Gamma_{22}$ consistently exhibit smaller values and similar magnitudes with opposite signs. This leads to a mutual cancellation, resulting in lower values for D^{M}_{12}. The contributions to D^{M}_{21} are given by $\Delta^{*}_{21}\Gamma_{11}$ and $\Delta^{*}_{22}\Gamma_{21}$. However, the composition-dependent trend of D^{M}_{21} is mainly governed by $\Delta^{*}_{22}\Gamma_{21}$. At x_1 = 0.10, $\Delta^{*}_{22}\Gamma_{21}$ undergoes a sign crossover from positive to negative and begins to decrease in its negative magnitude at

$x_1 > 0.15$. Generally, its magnitude primarily originates from Δ^*_{22} and follows the sign of Γ_{21}. The opposite behavior is observed for $\Delta^*_{21}\Gamma_{11}$, but its values are generally smaller in magnitude than those of $\Delta^*_{22}\Gamma_{21}$. The dominant contribution to D^M_{21} consistently comes from $\Delta^*_{21}\Gamma_{11}$. In the case of D^M_{22}, $\Delta^*_{21}\Gamma_{12}$ mainly determines the values up to $x_1 = 0.10$. After this point, both products generally contribute.

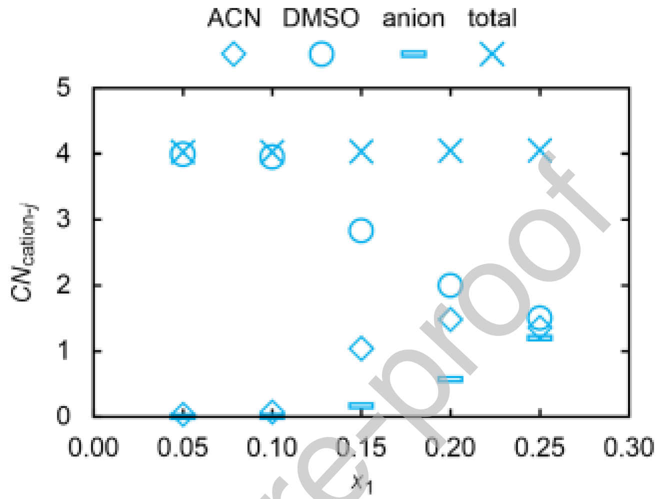

Figure 16. Coordination numbers CN_{ij} of ACN ($CN_{\text{cation-ACN}}$), DMC ($CN_{\text{cation-DMSO}}$) and anion ($CN_{\text{cation-anion}}$) around a cation, obtained from MD simulations for the mixtures containing [Li][NTf$_2$] dissolved in equimolar binary mixtures of ACN and DMSO as a function of amount fraction of the salt x_1 at T = 293.15 K. Error bars are not shown as they are smaller than the symbol sizes.

The solvation structure around the lithium cation was resolved in this study through the calculation of CNs, as explained under section 3.2. The results are depicted in Figure 16 for the mixtures at hand as a function of x_1 at T = 293.15 K. In line with previously studied mixtures with lower salt content, MD simulations results suggest that the cation solvation shell retains the four-site coordination structure over the investigated x_1 range. Up to x_1 = 0.10, only four DMSO molecules occupy the solvation shell surrounding the lithium cations. At x_1 = 0.15, ACN tends to replace one of the DMSO molecules in the solvation shell. When

x_1 is further increased to 0.20, the average number of DMSO molecules in the solvation shell decreases further. This decrease is accompanied by a further increase in ACN molecules in the solvation shell of the cations. An interpretation for this observation is given by the increasing number of lithium cations with increasing solute amount fraction, which requires more solvent molecules for the solvation of the ions. This also supports the increased preferential interactions between ACN and the salt molecules above $x_1 = 0.10$ as predicted by the negative Γ_{21} values. Meanwhile, $CN_{\text{cation-anion}}$ shows an increasing tendency which becomes most prominent when x_1 exceeds 0.20. This indicates the appearance of $[NTf_2]^-$ anions in the solvation shell of the cations, suggesting the formation of ion pairs. At higher x_1 values above 0.20, $CN_{\text{cation-DMSO}}$ maintains its steady decline, whereas the increasing trend of $CN_{\text{cation-ACN}}$ becomes reversed and begins to decrease. The latter shift aligns with the diminished magnitude of Γ_{21} observed when x_1 exceeds 0.20.

5 Conclusions

This work presents a systematic characterization of the diffusive mass transport in ternary electrolyte mixtures consisting of $[Li][NTf_2]$ dissolved in a binary mixture of ACN and DMC or ACN and DMSO over a broad range of temperatures and compositions. For this, mass diffusivities of the mixtures are determined by analyzing microscopic concentration fluctuations at macroscopic thermodynamic equilibrium using DLS and MD simulations. For the first time, this work proves that DLS is able to access both eigenvalues of the second-order Fick diffusion coefficient matrix $\boldsymbol{D}$ for a ternary electrolyte mixture and that the determined mass modes are related to hydrodynamic fluctuations. In the case of MD simulations, for the first time the complete methodology for accessing the components of $\boldsymbol{D}$ in the amount-averaged reference frame is presented.

For dilute electrolyte solutions with [Li][NTf_2] amount fractions of about 0.05, DLS accesses only one mass mode, which can be related to one of the two eigenvalues of $\boldsymbol{D}$. For the mixture based on ACN and DMC, the eigenvalue falls within the Fick diffusion coefficients of the binary subsystems of [Li][NTf_2] dissolved in either ACN or DMC. This indicates that the diffusion of the salt in the ternary mixture can be approximated by a pseudo-binary mixture of the lithium salt dissolved in a pseudo-pure solvent representing the binary solvent mixture. In the case of mixtures based on ACN and DMSO, it could be shown that both eigenvalues can be accessed simultaneously by DLS at $x_{\text{[Li][NTf2]}} > 0.15$. The simulation methodology and the employed force fields could be validated with the results from DLS and a good agreement between both datasets was found. The average absolute relative deviation (AARD) of the simulated eigenvalues from the experimental ones is about 21%.

Based on the good agreement between the results from simulations and experiments, MD simulations were further used to study the influence of intermolecular interactions on the diffusive mass transport. For this, the Fick diffusion coefficients were decoupled into their kinetic and thermodynamic contributions, expressed by MS diffusion coefficients and thermodynamic factors. Furthermore, also the fluid structure was analyzed by MD simulations by determining which solvent molecules are involved in the dissociation of the ions. This was done by calculating the coordination or solvation numbers. For mixtures based on ACN and DMC, it could be shown that ACN molecules can disproportionately be found in the solvation shell surrounding the lithium cations. This is also expressed by the negative off-diagonal element of the Fick diffusion coefficient matrix. For mixtures based on ACN and DMSO, on the other hand, DMSO is more involved in dissociating the ions in comparison to ACN. At low lithium concentrations, all four solvation sites in the solvation shell surrounding the lithium cation are occupied by DMSO. At $x_{\text{[Li][NTf2]}} > 0.15$, one of the four solvation sites is occupied by ACN, since not enough DMSO molecules are available in the mixture. Once $x_{[Li][NTf2]} > 0.20$, there are not enough solvent molecules available in the mixture to fully dissociate the lithium ions and $[NTf_2]^-$ anions start to appear in the solvation shell surrounding the cation. This represents the formation of ion pairs and clusters and is expected due to the approaching solubility limit. The formation of ion pairs and clusters also explains the slowing down of the Fickian diffusion process, as it hinders the free movement of the ions. With the insights gained from the present work, further investigation into how self- and cross-correlations influence the elements of $\boldsymbol{D}^{\mathrm{M}}$ can deepen our understanding of diffusive mass transport in ternary electrolyte mixtures. In the literature, MD simulations have been used to analyze their effects on conductivity in electrolyte mixtures, for example, by Kiyobayashi et al. [9]. However, to the best of our knowledge, the underlying procedures that enable investigation of the influence of these individual correlations on the Fick diffusion coefficients of electrolyte mixtures have yet to be established. Addressing these aspects represents an important direction for future research.

ASSOCIATED CONTENT

Supporting Information.

In the Supporting Information, the FF parameters and results from MD simulations are summarized. Furthermore, the results from DLS for the binary sub-system consisting of the two solvent molecules are summarized. Additionally, the primary measurement data from DLS, which are the temperature, pressure, composition, scattering angle, and the corresponding correlation function are summarized in the form of a data table (XLSX).

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Notes. The authors declare no competing financial interest.

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REFERNCES

[1] A. Brandt, A. Balducci, Theoretical and Practical Energy Limitations of Organic and Ionic Liquid-Based Electrolytes for High Voltage Electrochemical Double Layer Capacitors, J. Power Sources 250 (2014) 343–351. https://doi.org/10.1016/j.jpowsour.2013.10.147.

[2] R. Ye, D. Henkensmeier, S.J. Yoon, Z. Huang, D.K. Kim, Z. Chang, S. Kim, R. Chen, Redox Flow Batteries for Energy Storage: A Technology Review, J. Electrochem. En. Conv. Stor. 15 (2018) 1–21. https://doi.org/10.1115/1.4037248.

[3] C. Xiang, K.M. Papadantonakis, N.S. Lewis, Principles and Implementations of Electrolysis Systems for Water Splitting, Mater. Horiz. 3 (2016) 169–173. https://doi.org/10.1039/c6mh00016a.

[4] M. Li, C. Wang, Z. Chen, K. Xu, J. Lu, New Concepts in Electrolytes, Chem. Rev. (2020). https://doi.org/10.1021/acs.chemrev.9b00531.

[5] J. Smiatek, A. Heuer, M. Winter, Properties of Ion Complexes and Their Impact on Charge Transport in Organic Solvent-Based Electrolyte Solutions for Lithium Batteries: Insights from a Theoretical Perspective, Batteries 4 (2018) 62. https://doi.org/10.3390/batteries4040062.

[6] L.O. Valøen, J.N. Reimers, Transport Properties of $LiPF_6$-Based Li-Ion Battery Electrolytes, J. Electrochem. Soc. 152 (2005) A882–A891. https://doi.org/10.1149/1.1872737.

[7] K. Xu, Electrolytes and Interphases in Li-Ion Batteries and Beyond, Chem. Rev. 114 (2014) 11503–11618. https://doi.org/10.1021/cr500003w.

[8] F. Marchini, S. Herrera, W. Torres, A.Y. Tesio, F.J. Williams, E.J. Calvo, Surface Study of Lithium−Air Battery Oxygen Cathodes in Different Solvent−Electrolyte pairs, Langmuir 31 (2015) 9236–9245. https://doi.org/10.1021/acs.langmuir.5b02130.

[9] T. Kiyobayashi, S. Uchida, H. Ozaki, K. Kiyohara, Molecular Dynamics Simulation to Reveal the Transport Mechanism of $LiPF_6$ in Ethylene carbonate + Dimethylcarbonate Binary Solvent, J. Chem. Phys. 159 (2023) 74501. https://doi.org/10.1063/5.0164099.

[10] J. Newman, K.E. Thomas-Alyea, Electrochemical Systems, 3rd ed., John Wiley & Sons, New Jersey, USA, 2004.

[11] S.G. Stewart, J. Newman, The Use of UV/vis Absorption to Measure Diffusion Coefficients in $LiPF_6$ Electrolytic Solutions, J. Electrochem. Soc. 155 (2008) F13–F16. https://doi.org/10.1149/1.2801378.

[12] E.R. Logan, J.R. Dahn, Electrolyte Design for Fast-Charging Li-Ion Batteries, Trends Chem. 2 (2020) 354–366. https://doi.org/10.1016/j.trechm.2020.01.011.

[13] H. Lundgren, J. Scheers, M. Behm, G. Lindbergh, Characterization of the Mass-Transport Phenomena in a Superconcentrated LiTFSI:Acetonitrile Electrolyte, J. Electrochem. Soc. 162 (2015) A1334–A1340. https://doi.org/10.1149/2.0961507jes.

[14] R.A. Robinson, R.H. Stokes, Electrolyte Solutions: Second Revised Edition, Dover Publications, Incorporated, New York, USA, 2012.

[15] B.E. Poling, J.M. Prausnitz, J.P. O'connell, The Properties of Gases and Liquids, McGraw-Hill, New York, USA, 2001.

[16] D.G. Miller, Application of Irreversible Thermodynamics to Electrolyte Solutions. I. Determination of Ionic Transport Coefficients lij for Isothermal Vector Transport Processes in Binary Electrolyte Systems, J. Chem. Phys. 70 (1966) 2639–2659. https://doi.org/10.1021/j100880a033.

[17] T.W. Chapman, The Transport Properties of Concentrated Electrolytic Solutions, University of California Ph.D. Thesis (UCRL-17768), 1967.

[18] J.G. Kirkwood, R.L. Baldwin, P.J. Dunlop, L.J. Gosting, G. Kegeles, Flow Equations and Frames of Reference for Isothermal Diffusion in Liquids, J. Chem. Phys. 33 (1960) 1505–1513. https://doi.org/10.1063/1.1731433.

[19] D.G. Leaist, Electrolyte Diffusion in Multicomponent Solutions, Yale University dissertation (8110046), 1980.

[20] T. Klein, C.J. Kankanamge, T.M. Koller, M.H. Rausch, A.P. Fröba, M.J. Assael, W.A. Wakeham, G. Guevara-Carrion, J. Vrabec, Definitions and Preferred Symbols for Mass Diffusion Coefficients in Multicomponent Fluid Mixtures Including Electrolytes

(IUPAC Technical Report), Pure Appl. Chem. 97 (2025) 689–713. https://doi.org/10.1515/pac-2024-0251.

[21] L. Chen, D.G. Leaist, Multicomponent Taylor Dispersion Coefficients, J. Solut. Chem. 43 (2014) 2224–2237. https://doi.org/10.1007/s10953-014-0268-y.

[22] E.L. Cussler Jr, P.J. Dunlop, An Experimental Comparison of the Gouy and the Diaphragm Cell Methods for Studying Isothermal Ternary Diffusion, J. Phys. Chem. 70 (1966) 1880–1888.

[23] A. Revzin, New Procedure for Calculating the Four Diffusion Coefficients for Ternary Systems from Gouy Pptical Data. Application to Data for the System Potassium Bromide-Hydrobromic Acid-Water, J. Phys. Chem. 76 (1972) 3419–3429.

[24] P.J. Dunlop, Data for Diffusion in Concentrated Solution of the System NaCl–KCl–H_2O at 25°: A Test of the Onsager Reciprocal Relation for this Composition, J. Phys. Chem. 63 (1959) 612–615. https://doi.org/10.1021/j150574a040.

[25] E.L. Cussler, Multicomponent Diffusion, Elsevier Scientific Publishing Company, Amsterdam, Netherlands, 1976.

[26] B.J. Berne, R. Pecora, Dynamic Light Scattering: With Applications to Chemistry, Biology, and Physics, Dover Publications, New York, 2000.

[27] A. Bardow, On the Interpretation of Ternary Diffusion Measurements in Low-Molecular Weight Fluids by Dynamic Light Scattering, Fluid Phase Equilib. 251 (2007) 121–127. https://doi.org/10.1016/j.fluid.2006.11.006.

[28] L. Onsager, Reciprocal Relations in Irreversible Processes. I., Phys. Rev. 37 (1931) 405–426. https://doi.org/10.1103/PhysRev.37.405.

[29] L. Onsager, Reciprocal relations in irreversible processes. II, Phys. Rev. 38 (1931) 2265–2279. https://doi.org/10.1103/PhysRev.38.2265.

[30] D.A. Ivanov, T. Grossmann, J. Winkelmann, Comparison of Ternary Diffusion Coefficients Obtained from Dynamic Light Scattering and Taylor Dispersion, Fluid Phase Equilib. 228–229 (2005) 283–291. https://doi.org/https://doi.org/10.1016/j.fluid.2004.09.019.

[31] D.A. Ivanov, J. Winkelmann, Static and Dynamic Light Scattering Measurements Near the Critical Solution Point of a Ternary Liquid Mixture, Phys. Chem. Chem. Phys. 006 (2004) 3490–3499. https://doi.org/10.1039/B402165G.

[32] A. Heller, C. Giraudet, Z.A. Makrodimitri, M.S.H. Fleys, J. Chen, G.P. Van Der Laan, I.G. Economou, M.H. Rausch, A.P. Fröba, Diffusivities of Ternary Mixtures of *n*-Alkanes with Dissolved Gases by Dynamic Light Scattering, J. Phys. Chem. B 120

(2016) 10808–10823. https://doi.org/10.1021/acs.jpcb.6b08117.

[33] X. Liu, T.J.H. Vlugt, A. Bardow, Maxwell-Stefan Diffusivities in Binary Mixtures of Ionic Liquids with Dimethyl Dulfoxide (DMSO) and H_2O, J. Phys. Chem. B 115 (2011) 8506–8517. https://doi.org/10.1021/jp203026c.

[34] T. Klein, M. Piszko, M. Lang, J. Mehler, P.S. Schulz, M.H. Rausch, C. Giraudet, T.M. Koller, A.P. Fröba, Diffusivities in Binary Mixtures of [AMIM][NTf_2] Ionic Liquids with the Dissolved Gases H_2, He, N_2, CO, CO_2, or Kr Close to Infinite Dilution, J. Chem. Eng. Data 65 (2020) 4116–4129. https://doi.org/10.1021/acs.jced.0c00430.

[35] X. Liu, S.K. Schnell, J.-M. Simon, P. Krüger, D. Bedeaux, S. Kjelstrup, A. Bardow, T.J.H. Vlugt, Diffusion Coefficients from Molecular Dynamics Simulations in Binary and Ternary Mixtures, Int. J. Thermophys. 34 (2013) 1169–1196. https://doi.org/10.1007/s10765-013-1482-3.

[36] O. Borodin, M. Olguin, P. Ganesh, P.R.C. Kent, J.L. Allen, W.A. Henderson, Competitive Lithium Solvation of Linear and Cyclic Carbonates from Quantum Chemistry, Phys. Chem. Chem. Phys. 18 (2016) 164–175. https://doi.org/10.1039/C5CP05121E.

[37] C.J. Kankanamge, T. Zhan, M. Piszko, T. Klein, A.P. Fröba, Diffusion Coefficients in Binary Electrolyte Mixtures by Dynamic Light Scattering and Molecular Dynamics Simulations, Electrochim. Acta 462 (2023) 142637. https://doi.org/10.1016/j.electacta.2023.142637.

[38] C.J. Kankanamge, T. Zhan, Z. Zhang, T. Klein, A.P. Fröba, Influence of the Molecular Characteristics of the Solvent and the Solute Cation on the Diffusive Mass Transport in Binary Electrolyte Mixtures by Dynamic Light Scattering and Molecular Dynamics Simulations, J. Electrochem. Soc. 171 (2024) 110533. https://doi.org/10.1149/1945-7111/ad92df.

[39] C.J. Kankanamge, T. Klein, A.P. Fröba, Diffusion Coefficients in Electrolyte Mixtures─Influence of the Solute Characteristics, J. Chem. Eng. Data 70 (2025) 1300–1311. https://doi.org/10.1021/acs.jced.4c00671.

[40] C.J. Kankanamge, A.I. Diaz, T. Klein, A.P. Fröba, Prediciton of Fick Diffusion Coefficients in Electrolyte Mixtures, Int. J. Thermophys. 46 (2025) 142. https://doi.org/10.1007/s10765-025-03610-9.

[41] T. Janzen, J. Vrabec, Diffusion Coefficients of a Highly Nonideal Ternary Liquid Mixture: Cyclohexane–Toluene–Methanol, Ind. Eng. Chem. Res. 57 (2018) 16508–16517. https://doi.org/10.1021/acs.iecr.8b04385.

[42] S. Păez, G. Guevara-Carrion, H. Hasse, J. Vrabec, Mutual Diffusion in the Ternary Mixture of Water + Methanol + Ethanol and Its Binary Subsystems, Phys. Chem. Chem. Phys. 15 (2013) 3985–4001. https://doi.org/10.1039/C3CP43785J.

[43] X. Liu, A. Martín-Calvo, E. McGarrity, S.K. Schnell, S. Calero, J.M. Simon, D. Bedeaux, S. Kjelstrup, A. Bardow, T.J.H. Vlugt, Fick Diffusion Coefficients in Ternary Liquid Systems from Equilibrium Molecular Dynamics Simulations, Ind. Eng. Chem. Res. 51 (2012) 10247–10258. https://doi.org/10.1021/ie301009v.

[44] S. Kjelstrup, A.F. Gunnarshaug, Ø. Gullbrekken, S.K. Schnell, A. Lervik, Transport Coefficients for Ion and Solvent Coupling. The Case of the Lithium-Ion Battery Electrolyte, J. Chem. Phys. 159 (2023) 34104. https://doi.org/10.1063/5.0158623.

[45] A. Fick, Über Diffusion, Ann. Phys. 170 (1855) 59–86. https://doi.org/doi.org/10.1002/andp.18551700105.

[46] R. Taylor, R. Krishna, Multicomponent Mass Transfer, John Wiley & Sons, New York, 1993.

[47] J.C. Maxwell, On the Dynamical Theory of Gases, Phil. Trans. R. Soc. London I 157 (1867) 49–88.

[48] J. Stefan, Über das Gleichgewicht und die Bewegung, insbesondere die Diffusion von Gasgemengen, Sitzungsberichte Der Math. Cl. Der Kais. Akad. Der Wissenschaften Wien, 2te Abteilung 63 (1871) 63–124.

[49] G. Guevara-Carrion, R. Fingerhut, J. Vrabec, Fick Diffusion Coefficient Matrix of a Quaternary Liquid Mixture by Molecular Dynamics, J. Phys. Chem. B 124 (2020) 4527–4535. https://doi.org/10.1021/acs.jpcb.0c01625.

[50] L.A. Woolf, D.G. Miller, L.J. Gosting, Isothermal Diffusion Measurements on the System H_2O-Glycine-KCl at 25^o; Tests of the Onsager Reciprocal Relation, J. Am. Chem. Soc. 84 (1962) 317–331. https://doi.org/10.1021/ja00862a001.

[51] J.M. Ortiz de Zárate, J. V Sengers, Frame-Invariant Fick Diffusion Matrices of Multicomponent Fluid Mixtures, Phys. Chem. Chem. Phys. 22 (2020) 17597–17604. https://doi.org/10.1039/D0CP01110J.

[52] D.G. Miller, Some Comments on Multicomponent Diffusion: Negative Main Term Diffusion Coefficients, Second Law Constraints, Solvent Choices, and Reference Frame Transformations, J. Chem. Phys. 90 (1986) 1509–1519. https://doi.org/10.1021/j100399a010.

[53] H.T.J. Cullinan, Analysis of Flux Equations of Multicomponent Diffusion, Ind. Eng. Chem. Fundamen. 4 (1965) 133–139. https://doi.org/10.1021/i160014a005.

[54] D.N. Bennion, Phenomena at a Gas-Electrode-Electrolyte Interface, University of California dissertation, 1957.

[55] M. Piszko, C.J. Kankanamge, M.H. Rausch, T. Klein, A.P. Fröba, Dynamic Light Scattering for Studying Mutual Diffusion Coefficients in Electrolyte Systems Comprised Entirely of Ions, J. Electrochem. Soc. 167 (2020) 133502. https://doi.org/10.1149/1945-7111/abb97d.

[56] Y. Zhang, Y. Zhang, M.J. McCready, E.J. Maginn, Evaluation and Refinement of the General AMBER Force Field for Nineteen Pure Organic Electrolyte Solvents, J. Chem. Eng. Data 63 (2018) 3488–3502. https://doi.org/10.1021/acs.jced.8b00382.

[57] M.H. Rausch, L. Hopf, A. Heller, A. Leipertz, A.P. Fröba, Binary Diffusion Coefficients for Mixtures of Ionic Liquids [EMIM][$N(CN)_2$], [EMIM][NTf_2], and [HMIM][NTf_2] with Acetone and Ethanol by Dynamic Light Scattering (DLS), J. Phys. Chem. B 117 (2013) 2429–2437. https://doi.org/10.1021/jp306116h.

[58] T.M. Koller, M.H. Rausch, A.P. Fröba, Dynamic Light Scattering for the Measurement of Transport Properties of Fluids, Int. J. Thermophys. 45 (2024) 57. https://doi.org/10.1007/s10765-024-03344-0.

[59] A.P. Fröba, Dynamic Light Scattering (DLS) for the Characterization of Working Fluids in Chemical and Energy Engineering, Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, Germany, 2009.

[60] D. Mombeini, H.P. Blabus, M.H. Rausch, T. Klein, T.M. Koller, C.J. Kankanamge, M. Geißelbrecht, P. Wasserscheid, A.P. Fröba, Fick Diffusivity, Thermal Diffusivity, Thermal Conductivity, Density, and Mixture Composition of Binary Mixtures of γ-Butyrolactone and 1,4-Butanediol by Light Scattering Techniques and Conventional Methods, Int. J. Thermophys. 47 (2026) 28. https://doi.org/10.1007/s10765-025-03694-3.

[61] R. Pecora, Dynamic Light Scattering: Applications of Photon Correlation Spectroscopy, Springer, New York, USA, 1985.

[62] D.G. Leaist, L. Hao, Comparison of Diffusion Coefficients of Multicomponent Solutions from Light Scattering and Macroscopic Gradient Techniques: Sodium Dodecyl Sulfate Micelles in Squeous Salt Solutions, J. Phys. Chem. 97 (1993) 7763–7768. https://doi.org/10.1021/j100131a054.

[63] R.D. Mountain, J.M. Deutch, Light Scattering from Binary Solutions, J. Chem. Phys. 50 (1969) 1103–1108. https://doi.org/10.1063/1.1671163.

[64] M. Piszko, T. Schaible, C. Bonten, A.P. Fröba, Mutual and Thermal Diffusivities in

Polystyrene Melts with Dissolved Nitrogen by Dynamic Light Scattering, Macromolecules 54 (2021) 5662–5672. https://doi.org/10.1021/acs.macromol.1c00819.

[65] C.S. Johnson, D.A. Gabriel, Laser Light Scattering, Dover Publications, Inc., New York, USA, 2015.

[66] P.K. Chittem, N.S. Gálvez, P. Malgaretti, J. Harting, N. Vogel, P.S. Schmidt, A.P. Fröba, Determination of Particle Diffusion in a Confined Fluid Layer by Dynamic Light Scattering, Colloids Surf., A 737 (2026) 139892. https://doi.org/https://doi.org/10.1016/j.colsurfa.2026.139892.

[67] W. Wu, J. Cui, U. Sultan, L. Gromotka, P. Malgaretti, C. Damm, J. Harting, N. Vogel, W. Peukert, A. Inayat, A.P. Fröba, Diffusion of Gold Nanoparticles in Porous Silica Monoliths Determined by Dynamic Light Scattering, J. Colloid Interface Sci. 641 (2023) 251–264. https://doi.org/https://doi.org/10.1016/j.jcis.2023.03.045.

[68] T. Klein, W. Wu, M.H. Rausch, C. Giraudet, T.M. Koller, A.P. Fröba, Influence of Liquid Structure on Fickian Diffusion in Binary Mixtures of *n*-Hexane and Carbon Dioxide Probed by Dynamic Light Scattering, Raman Spectroscopy, and Molecular Dynamics Simulations, J. Phys. Chem. B 122 (2018) 7122–7133. https://doi.org/10.1021/acs.jpcb.8b03568.

[69] J.M.O. De Zarate, J. V Sengers, Hydrodynamic Fluctuations in Fluids and Fluid Mixtures, Elsevier , Amsterdam, 2006.

[70] M.P. Allen, D.J. Tildesley, Computer Simulation of Liquids, Oxford University Press, Oxford, UK, 2017.

[71] D.C. Rapaport, The Art of Molecular Dynamics Simulation, Cambridge University Press, Cambridge, UK, 2004. https://doi.org/10.2277/0521825687.

[72] G. Raabe, Molecular Simulation Studies on Thermophysical Properties: With Application to Working Fluids, Springer, Singapore, 2017. https://doi.org/10.1007/978-981-10-3545-6.

[73] X. Liu, T.J.H. Vlugt, A. Bardow, Maxwell-Stefan diffusivities in liquid mixtures: Using molecular dynamics for testing model predictions, Fluid Phase Equilib. 301 (2011) 110–117. https://doi.org/10.1016/j.fluid.2010.11.019.

[74] D. Frenkel, B. Smit, Understanding Molecular Simulation: From Algorithms to Applications, Academic Press, San Diego, USA, 2002.

[75] Lindahl, Abraham, Hess, van der Spoel, GROMACS 2021.5 Source code, (2022). https://doi.org/10.5281/zenodo.5850051.

[76] W.L. Jorgensen, D.S. Maxwell, J. Tirado-Rives, Development and Testing of the OPLS All-Atom Force Field on Conformational Energetics and Properties of Organic Liquids, J. Am. Chem. Soc. 118 (1996) 11225–11236. https://doi.org/10.1021/ja9621760.

[77] M.J. Abraham, T. Murtola, R. Schulz, S. Páll, J.C. Smith, B. Hess, E. Lindah, Gromacs: High Performance Molecular Simulations through Multi-Level Parallelism from Laptops to Supercomputers, SoftwareX 1–2 (2015) 19–25. https://doi.org/10.1016/j.softx.2015.06.001.

[78] T. Darden, D. York, L. Pedersen, Particle Mesh Ewald: An N·log(N) Method for Ewald Sums in Large Systems, J. Chem. Phys. 98 (1993) 10089–10092. https://doi.org/10.1063/1.464397.

[79] B. Hess, H. Bekker, H.J.C. Berendsen, J.G.E.M. Fraaije, LINCS: A Linear Constraint Solver for Molecular Simulations, J. Comput. Chem. 18 (1997) 1463–1472. https://doi.org/10.1002/(SICI)1096-987X(199709)18:12<1463::AID-JCC4>3.0.CO;2-H.

[80] G. Bussi, D. Donadio, M. Parrinello, Canonical sampling through velocity rescaling, J. Chem. Phys. 126 (2007) 14101. https://doi.org/10.1063/1.2408420.

[81] S. Nosé, M.L. Klein, Constant Pressure Molecular Dynamics for Molecular Systems, Mol. Phys. 50 (1983) 1055–1076. https://doi.org/10.1080/00268978300102851.

[82] M. Parrinello, A. Rahman, Polymorphic Transitions in Single Crystals: A New Molecular Dynamics Method, J. Appl. Phys. 52 (1981) 7182–7190. https://doi.org/10.1063/1.328693.

[83] L. Martínez, R. Andrade, E.G. Birgin, J.M. Martínez, PACKMOL: A Package for Building Initial Configurations for Molecular Dynamics Simulations, J Comput. Chem. 30 (2009) 2157–2164.

[84] M.L.P. Price, D. Ostrovsky, W.L. Jorgensen, Gas-Phase and Liquid-State Properties of Esters, Nitriles, and Nitro Compounds with the OPLS-AA Force Field, J. Comput. Chem. 22 (2001) 1340–1352. https://doi.org/10.1002/jcc.1092.

[85] J.-C. Soetens, C. Millot, B. Maigret, Molecular Dynamics Simulation of $Li^+BF_4^-$ in Ethylene Carbonate, Propylene Carbonate, and Dimethyl Carbonate Solvents, J. Phys. Chem. A 102 (1998) 1055–1061. https://doi.org/10.1021/jp972457+.

[86] X. Liu, S.K. Schnell, J.-M. Simon, D. Bedeaux, S. Kjelstrup, A. Bardow, T.J.H. Vlugt, Fick Diffusion Coefficients of Liquid Mixtures Directly Obtained From Equilibrium Molecular Dynamics, J. Phys. Chem. B 115 (2011) 12921–12929.

https://doi.org/10.1021/jp208360s.

[87] R. Krishna, J.M. Van Baten, The Darken Relation for Multicomponent Diffusion in Liquid Mixtures of Linear Alkanes: An Investigation Using Molecular Dynamics (MD) Simulations, Ind. Eng. Chem. Res. 44 (2005) 6939–6947. https://doi.org/10.1021/ie050146c.

[88] S.K. Schnell, P. Englebienne, J.M. Simon, P. Krüger, S.P. Balaji, S. Kjelstrup, D. Bedeaux, A. Bardow, T.J.H. Vlugt, How to apply the Kirkwood-Buff theory to individual species in salt solutions, Chem. Phys. Lett. 582 (2013) 154–157. https://doi.org/10.1016/j.cplett.2013.07.043.

[89] A. Ben-Naim, Molecular Theory of Solutions, Oxford University Press, Oxford, 2006. https://doi.org/10.1017/CBO9781107415324.004.

[90] R. Fingerhut, J. Vrabec, Kirkwood-Buff Integration: A Promising Route to Entropic Properties?, Fluid Phase Equilib. 485 (2019) 270–281. https://doi.org/10.1016/j.fluid.2018.12.015.

[91] P. Krüger, S.K. Schnell, D. Bedeaux, S. Kjelstrup, T.J.H. Vlugt, J.M. Simon, Kirkwood-Buff Integrals for Finite Volumes, J. Phys. Chem. Lett. 4 (2013) 235–238. https://doi.org/10.1021/jz301992u.

[92] P. Krüger, T.J.H. Vlugt, Size and Shape Dependence of Finite-Volume Kirkwood−Buff Integrals, Phys. Rev. E 97 (2018) 1–5. https://doi.org/10.1103/PhysRevE.97.051301.

[93] N. Dawass, P. Krüger, S.K. Schnell, O.A. Moultos, I.G. Economou, T.J.H. Vlugt, J.-M. Simon, Kirkwood-Buff Integrals Using Molecular Simulation: Estimation of Surface Effects, Nanomaterials 10 (2020). https://doi.org/10.3390/nano10040771.

[94] S.K. Schnell, X. Liu, J.-M. Simon, A. Bardow, D. Bedeaux, T.J.H. Vlugt, S. Kjelstrup, Calculating Thermodynamic Properties from Fluctuations at Small Scales, J. Phys. Chem. B 115 (2011) 10911–10918. https://doi.org/10.1021/jp204347p.

[95] P. Ganguly, N.F.A. van der Vegt, Convergence of Sampling Kirkwood−Buff Integrals of Aqueous Solutions with Molecular Dynamics Simulations, J. Chem. Theory Comput. 9 (2013) 1347–1355. https://doi.org/10.1021/ct301017q.

[96] R. Chitra, P.E. Smith, Preferential Interactions of Cosolvents with Hydrophobic Solutes, J. Phys. Chem. B 105 (2001) 11513–11522. https://doi.org/10.1021/jp012354y.

[97] R. Chitra, P.E. Smith, Molecular Association in Solution: A Kirkwood-Buff Analysis of Sodium Chloride, Ammonium Sulfate, Guanidinium Chloride, Urea, and 2,2,2-

Trifluoroethanol in Water, J. Phys. Chem. B 106 (2002) 1491–1500. https://doi.org/10.1021/jp011462h.

[98] M.B. Gee, N.R. Cox, Y. Jiao, N. Bentenitis, S. Weerasinghe, P.E. Smith, A Kirkwood−Buff Derived Force Field for Aqueous Alkali Halides, J. Chem. Theory Comput. 7 (2011) 1369–1380. https://doi.org/10.1021/ct100517z.

[99] Y. Zhang, N.H.C. Lewis, J. Mars, G. Wan, N.J. Weadock, C.J. Takacs, M.R. Lukatskaya, H.G. Steinrück, M.F. Toney, A. Tokmakoff, E.J. Maginn, Water-in-Salt LiTFSI Aqueous Electrolytes. 1. Liquid Structure from Combined Molecular Dynamics Simulation and Experimental Studies, J. Phys. Chem. B 125 (2021) 4501–4513. https://doi.org/10.1021/acs.jpcb.1c02189.

[100] T. Hou, C.W. Monroe, Composition-Dependent Thermodynamic and Mass-Transport Characterization of Lithium Hexafluorophosphate in Propylene Carbonate, Electrochim. Acta 332 (2020) 135085. https://doi.org/10.1016/j.electacta.2019.135085.

[101] N. Mozhzhukhina, M.P. Longinotti, H.R. Corti, E.J. Calvo, A Conductivity Study of Preferential Solvation of Lithium Ion in Acetonitrile-Dimethyl Sulfoxide Mixtures, Electrochim. Acta 154 (2015) 456–461. https://doi.org/https://doi.org/10.1016/j.electacta.2014.12.022.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Graphical abstract